

Machine-Provable Diagnosis of Biological Sequence Information via Karnaugh Map Boolean Minimisation: From Lean 4 Formal Verification to Cross-Protein Coevolutionary Contact Prediction

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Abstract

The representation of biological sequences as Karnaugh maps through Gray-code encoding establishes a bridge between the digital-logic design formalism of switching theory and the statistical mechanics of protein coevolution. In this work we demonstrate that the entire analytical chain—from amino-acid integer encoding through Gray-code transformation to Karnaugh-map construction, Quine–McCluskey minimisation, and prime-implicant extraction—can be made machine-provable in the Lean 4 theorem prover, and that the resulting Boolean circuits carry genuine structural information when applied to multiple sequence alignments of protein families. We formalise 118 theorems covering Gray-code properties, k -mer indexing, amino-acid encodings, contact-map topology, and implicant-cover semantics; we then apply the verified pipeline to 150 protein domains from the PSICOV benchmark comprising over 1.6 million labelled residue-pair observations against native experimental structures. Our results show that Quine–McCluskey-minimised contact circuits transfer across proteins at a mean top- $L/5$ precision of 0.179 ($5.1\times$ random enrichment), that Gray-code adjacency is enriched among native-contact pairs ($1.178\times$, $p = 1.1 \times 10^{-77}$), and that a rank-fusion hybrid combining mean-field DCA scores with the Karnaugh-map cell-rate prior achieves 0.204 top- $L/5$ precision, exceeding both individual methods ($p < 10^{-4}$). We additionally demonstrate that flipped (forbidden-combination) circuits transfer across proteins with 98.3% specificity and that conserved column pairs are enriched for native contacts by $3.2\times$. All software, data, and formal proofs are publicly available.

1 Introduction

The prediction of residue–residue contacts from multiple sequence alignments (MSAs) has emerged as one of the most consequential applications of evolutionary sequence analysis, underpinning modern protein structure prediction pipelines [1] and enabling the determination of protein folds from covariation signal alone [2]. The dominant paradigm, Direct Coupling Analysis (DCA), models the joint probability distribution of amino-acid sequences through a Potts Hamiltonian whose coupling parameters are inferred by inverting a covariance matrix computed from deep alignments [3]. While methods based on sparse inverse covariance estimation [4], pseudo-likelihood maximisation [5], and related formulations have achieved remarkable precision at the top- $L/5$ long-range contact level, the mathematical objects that these methods produce—a dense matrix of coupling constants indexed by residue pairs and amino-acid identities—remain largely opaque to formal analysis. In particular, no prior work has established machine-checkable proofs that the

encoding layer through which biological sequences enter the statistical machinery preserves the structural properties that the downstream inference requires.

This paper addresses that gap by constructing a formally verified bridge between digital-logic design theory and protein coevolution analysis. Our central observation is that the Karnaugh map, a rectangular grid representation of Boolean functions introduced by Karnaugh in 1953 [6], provides a natural embedding of aligned biological sequences when combined with reflected binary Gray-code ordering of the amino-acid alphabet. Under this embedding, adjacent cells on the map correspond to pairs of residues that differ in exactly one bit position of their Gray codes, which in turn correspond to physicochemically similar amino acids under an appropriate ordering of the twenty standard residues [7]. The Quine–McCluskey algorithm [8, 9] then extracts minimal sum-of-products (SOP) expressions from thresholded frequency maps, producing sets of prime implicants that we interpret as coevolutionary rules of the form “if residue a appears at position i and residue b at position j , then positions i and j are coevolutionarily constrained.”

The contribution of this work is threefold. First, we formalise the entire analytical chain in the Lean 4 theorem prover [10], establishing 118 machine-checked theorems that cover Gray-code bijectivity and adjacency, k -mer indexing injectivity, amino-acid encoding properties, contact-map symmetry and irreflexivity, and implicant-cover soundness and completeness. Second, we apply the verified pipeline to 150 protein domains from the PSICOV benchmark [4], comprising over 1.6 million labelled residue-pair observations against native experimental structures, and demonstrate that the resulting Boolean circuits transfer across proteins: a universal contact circuit trained on approximately one hundred proteins predicts native contacts on held-out proteins at a mean top- $L/5$ precision of 0.179, corresponding to $5.1\times$ enrichment over random expectation. Third, we show that the Karnaugh-map cell-rate prior, when fused with mean-field DCA scores via simple rank aggregation, yields a hybrid predictor that exceeds both individual methods (0.204 vs. 0.169 for DCA alone and 0.124 for the circuit alone; $p < 10^{-4}$ by Wilcoxon signed-rank test), providing the first demonstration that Boolean-minimisation-derived features carry orthogonal structural information not present in continuous coupling scores.

We further report honest negative results. An attention-style “tension” mechanism, which normalises each position’s mutual-information profile by its strongest entropic partner, does not improve contact prediction beyond what consensus identity features already provide. Iterative direct information computed from our mean-field implementation remains unreliable on families with many fully conserved columns due to pseudo-inverse conditioning, and we therefore adopt the APC-corrected Frobenius norm as the primary DCA score. Finally, we assess reconstruction readiness against literature thresholds and find that current circuit-level signal, while significantly above random expectation, falls far short of the precision required for template-free folding; the circuits are best understood as a compressed annotation layer that identifies specific coevolutionary constraints rather than a replacement for continuous physics-based scoring.

The paper is organised as follows. Section 2 introduces the Gray-code encoding framework for nucleotide and amino-acid alphabets. Section 3 describes Karnaugh-map construction and Quine–McCluskey minimisation with don’t-care conditions. Section 4 reviews the information-theoretic measures used throughout. Section 5 summarises Direct Coupling Analysis. Section 6 presents the Lean 4 formal verification. Sections 7–17 describe the dataset, pipeline architecture, and validation framework. Sections 19–29 present all positive and negative results. Section 32 discusses findings in context, and Section 33 concludes.

2 Gray-Code Encoding of Biological Alphabets

2.1 Reflected Binary Gray Codes

A reflected binary Gray code assigns to each non-negative integer n the value $g(n) = n \oplus (n \gg 1)$, where $\oplus$ denotes bitwise exclusive-or and $\gg$ denotes arithmetic right shift [11]. The defining property of this encoding is that consecutive integers map to codewords that differ in exactly one bit position, a property that follows from the identity $g(n) \oplus g(n+1) = 2^{\nu_2(n+1)}$, where $\nu_2(m)$ is the two-adic valuation of m . Over a fixed width of w bits, the mapping $g : \{0, \dots, 2^w - 1\} \rightarrow \{0, \dots, 2^w - 1\}$ is an involution (that is, applying it twice recovers the original value), and its image forms a Hamiltonian cycle on the w -dimensional hypercube Q_w .

The Hamming distance between two Gray-encoded values a and b is defined as

$$d_G(a, b) = \text{popcount}(g(a) \oplus g(b)), \quad (1)$$

where $\text{popcount}(x)$ counts the number of set bits in the binary representation of x . This distance inherits the metric properties of the underlying Hamming metric on $\{0, 1\}^w$ and satisfies the symmetry condition $d_G(a, b) = d_G(b, a)$ for all $a, b < 2^w$.

2.2 DNA and RNA Encoding

For nucleotide sequences, we adopt the standard two-bit encoding in which adenine (A), cytosine (C), guanine (G), and thymine/uracil (T/U) are assigned the binary codes $00_2 = 0$, $01_2 = 1$, $11_2 = 3$, and $10_2 = 2$, respectively. Under this assignment, the Gray-code transformation maps A to 0, C to 1, G to 2, and T to 3, and the transition–transversion classification of point mutations acquires a direct interpretation in terms of Gray-code adjacency: transitions (purine $\leftrightarrow$ purine or pyrimidine $\leftrightarrow$ pyrimidine) correspond to single bit-flips, while transversions correspond to two-bit flips. This property was formalised in Lean 4 as part of the verification framework described in Section 6.

2.3 Amino-Acid Encoding

For amino-acid sequences, we extend the encoding from four nucleotides to twenty canonical residues by assigning each amino acid a five-bit code under the He 2012 physicochemical ordering [7]. The ordering groups the twenty residues into seven classes based on hydrophobicity, charge, and aromaticity:

Group	Residues	He indices
Aliphatic	A, I, L, V	0–3
Aromatic	F, Y, W	4–6
Sulfur	M, C	7–8
Structure	P, G	9–10
Polar	S, T, N, Q	11–14
Negative	D, E	15–16
Positive	H, K, R	17–19

Under this ordering, the integer code assigned to residue a_i at position i is simply its He index, denoted $\text{code}(a_i) \in \{0, \dots, 19\}$, and the corresponding five-bit Gray code is $g(\text{code}(a_i))$. The twelve unused five-bit codewords ($20 \leq x \leq 31$) are designated as don't-care cells during Karnaugh-map minimisation, a design choice whose formal justification is given in Section 3.

The key property that makes this encoding suitable for Karnaugh-map analysis is that within-group pairs of residues tend to have small Gray-Hamming distances. Specifically,

among the $\binom{20}{2} = 190$ unordered pairs of canonical residues, 40 pairs (ordered count: 80) lie at Gray distance 1, and these are enriched for within-group substitutions relative to what would be expected under random assignment of codewords to residues. This enrichment was verified exhaustively by native computation in Lean 4 and is stated as the theorem `cc_he_dist1_ordered_eq_80` in the `ContactCircuits` module.

2.4 Gap Handling

Alignment gap characters (‘-’) and ambiguous residue calls (‘X’) are mapped to a twenty-first state with code value 20. This state is excluded from all statistical calculations (entropy, mutual information, coupling estimation) but is retained in the position arrays so that column identity is preserved across all sequences. The retention of gaps as state 20, rather than their deletion, constitutes the first of two critical corrections applied to our pipeline; the deletion of gaps causes different sequences to contribute to different raw positions at any given column index, producing spurious variable-position calls and inflated mutual-information values.

3 Karnaugh Maps and Quine–McCluskey Minimisation

3.1 Karnaugh Maps for Residue-Pair Statistics

For a pair of alignment columns (i, j) with $i < j$, we define a ternary Karnaugh map K_{ij} as a 32×32 grid whose rows and columns are indexed by the five-bit Gray codes of the consensus residues at positions i and j , respectively. Each cell $(a, b) \in \{0, \dots, 31\}^2$ of the map takes one of three values:

$$K_{ij}(a, b) = \begin{cases} +1 & \text{if the residue pair } (a, b) \text{ is an observed mutation,} \\ -1 & \text{if } (a, b) \text{ is the reference (majority) pair,} \\ 0 & \text{if } (a, b) \text{ has never been observed.} \end{cases} \quad (2)$$

Rows and columns with indices 20–31 correspond to unused Gray codewords and are assigned don’t-care status (-1), which permits the Quine–McCluskey algorithm to form larger implicants by merging on-set cells through these padding regions. This padding step constitutes the second critical correction applied to our pipeline; without it, a 20×20 map fed to a minimiser expecting $4^k = 2^{10} = 1024$ cells would silently wrap indices 256–399 onto indices 0–143, corrupting minterm labels and producing phantom rules.

3.2 Ternary Truth Tables

For computational purposes, each Karnaugh map is flattened into a one-dimensional truth table $\tau : \{0, \dots, 1023\} \rightarrow \{-1, 0, 1\}$ where the cell index c encodes both residue identities in MSB-first bit order:

$$c = 32 \cdot r + s, \quad r = g(\text{code}(a_i)), \quad s = g(\text{code}(a_j)), \quad (3)$$

and the bit at position b ($b = 0, \dots, 9$) of cell c is extracted as $(c \gg (9 - b)) \& 1$. An *implicant* over w variables is a pair $(\text{values}, \text{mask}) \in \{0, 1\}^w \times \{0, 1\}^w$ where a mask bit equal to 1 marks the corresponding variable position as don’t-care. An implicant *matches* a cell c when every constrained (mask = 0) position agrees with the corresponding bit of c :

$$\text{matches}(v, m, c) \iff ((v \oplus c) \& (\sim m)) = 0, \quad (4)$$

where $\sim m$ denotes the bitwise complement of m within the w -bit field.

3.3 Quine–McCluskey Algorithm

The Quine–McCluskey algorithm [8, 9] finds all prime implicants of a Boolean function by exhaustive pairwise merging of minterms that differ in exactly one bit position. Our implementation extends the classical formulation to handle ternary truth tables by including don’t-care cells in the initial minterm set alongside on-set cells, thereby allowing prime implicants to cover don’t-care regions without requiring them to do so. The merge phase groups minterms by popcount of their fixed bits and restricts comparisons to adjacent popcount groups, reducing the worst-case complexity from $O(m^2)$ to $O(m \cdot |\text{groups}|)$ for m minterms.

After all merges are exhausted, the surviving implicants that cannot be merged further are the *prime implicants*. Among these, a prime implicant is *essential* if it covers at least one on-set cell that no other prime implicant covers. A minimum cover is then constructed greedily from the essential prime implicants plus additional non-essential primes selected to cover any remaining uncovered on-set cells. Two properties of this procedure are machine-verified in Lean 4:

Theorem 3.1 (Cover Completeness). *Every on-set cell of a labeled Karnaugh map is matched by at least one prime implicant in the Quine–McCluskey output.*

Theorem 3.2 (Cover Soundness). *No prime implicant matches any off-set cell of a labeled Karnaugh map.*

Together, these two properties guarantee that the SOP expression produced by the minimiser neither omits nor overapproximates the coevolutionary signal, provided the input truth table correctly distinguishes observed mutations from unobserved combinations and from reference-pair don’t-cares.

3.4 Flipped Maps and Negative Selection

In addition to the positive (mutation-detecting) map described above, we construct a *flipped* map for each candidate pair by assigning on-set status to cells that have never been observed across all sequences in the alignment, off-set status to cells that have been observed, and don’t-care status to reference pairs. Prime implicants of the flipped map encode *forbidden* residue-pair combinations—pairs whose joint absence across all sequenced homologues suggests destabilising interactions or steric incompatibility. Together, the positive and flipped maps partition the candidate space into three disjoint regions: coevolutionary constraints, negative-selection boundaries, and uninformative gaps.

4 Information-Theoretic Measures

4.1 Shannon Entropy

For an alignment column i containing N sequences, the Shannon entropy [12] quantifies the evolutionary variability at that position:

$$H(i) = - \sum_{a=1}^{20} P_i(a) \log_2 P_i(a), \quad (5)$$

where $P_i(a)$ denotes the observed frequency of amino acid a at column i , computed after excluding gap characters (state 20). The entropy ranges from $H = 0$ (fully conserved column) to $H = \log_2(20) \approx 4.32$ bits (maximally variable). We classify a column as *variable* if $H(i) > 0.3$, a threshold chosen empirically to capture positions where at least two or three different amino acids appear with appreciable frequency.

4.2 Mutual Information

For a pair of alignment columns (i, j) , the mutual information measures the statistical dependence between the amino-acid distributions at those columns:

$$\text{MI}(i, j) = \sum_{a=1}^{20} \sum_{b=1}^{20} P_{ij}(a, b) \log_2 \frac{P_{ij}(a, b)}{P_i(a) P_j(b)}, \quad (6)$$

where $P_{ij}(a, b)$ is the joint frequency of observing residue a at column i and residue b at column j . MI is measured in bits and equals zero when the two positions are statistically independent. We compute two variants: the *full MI*, which counts all observations including the reference (majority-residue) pair, and the *mutation-only MI*, which excludes the reference pair to isolate mutation-driven covariation from conservation-driven signal.

Raw MI suffers from two well-known biases: phylogenetic inertia inflates correlations between sequences sharing recent common ancestry, and random sampling noise produces nonzero baseline values even for independent positions. The Average Product Correction (APC) of Dunn et al.[13] removes much of this bias by subtracting the product of row and column means from the MI matrix:

$$\text{MI}_{\text{APC}}(i, j) = \text{MI}(i, j) - \frac{\bar{\text{MI}}(i, \cdot) \bar{\text{MI}}(\cdot, j)}{\bar{\text{MI}}}, \quad (7)$$

where $\bar{\text{MI}}(i, \cdot)$ is the mean MI over all partners of position i and $\bar{\text{MI}}$ is the grand mean.

4.3 Perplexity Ratio

While MI measures total statistical dependence, it does not distinguish between many weakly contributing states and a few strongly determined ones. To capture this distinction, we define the *perplexity ratio* for a pair (i, j) as

$$r(i, j) = \frac{\text{PP}(j)}{\frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \text{PP}(j \mid i = a)}, \quad (8)$$

where $\text{PP}(j) = 2^{H(j)}$ is the marginal perplexity of column j , $\text{PP}(j \mid i = a) = 2^{H(j \mid i = a)}$ is the conditional perplexity given that column i holds residue a , and $\mathcal{A}$ is the set of residues at column i with at least five observations. A ratio exceeding unity indicates that knowing the residue at position i reduces the effective number of choices at position j . In our PSICOV150 evaluation, the perplexity ratio achieved the highest Spearman correlation with native-contact fraction among all sequence-derived metrics tested ($\rho = 0.670$, $p = 0.006$).

4.4 Coupling Constants

The coupling constant for a residue pair (a, b) at positions (i, j) is defined as the log-odds ratio

$$J_{ij}(a, b) = \ln \frac{P_{ij}(a, b)}{P_i(a) P_j(b)}, \quad (9)$$

which equals zero when the pair occurs at its independent (marginal-product) frequency. Positive values indicate coevolutionary enrichment; negative values indicate anti-correlation or negative selection. The sigmoid transform $\sigma(J) = (1 + e^{-J})^{-1}$ maps couplings to probabilities in $[0, 1]$.

5 Direct Coupling Analysis

5.1 The Potts Model

Direct Coupling Analysis (DCA) models the probability distribution of amino-acid sequences through a pairwise Potts Hamiltonian [2, 3]:

$$P(a_1, \dots, a_L) \propto \exp \left\{ \sum_{i < j} e_{ij}(a_i, a_j) + \sum_i h_i(a_i) \right\}, \quad (10)$$

where $e_{ij}(a, b)$ denotes the coupling between residues a and b at positions i and j , and $h_i(a)$ is a single-site bias field. The central computational challenge is the inverse problem: given an MSA, estimate the couplings e_{ij} that reproduce the observed single-site and pairwise frequencies.

5.2 Mean-Field Approximation

In the mean-field (MF) approximation [3], the couplings are obtained by inverting the connected-correlation (covariance) matrix:

$$e_{ij}(a, b) \approx -(C^{-1})_{(i,a),(j,b)}, \quad (11)$$

where C has dimension $(q-1)L \times (q-1)L$ with $q = 21$ states (20 canonical amino acids plus one gap state), and the gap state is removed as a gauge-fixing condition to eliminate the rank deficiency arising from the normalisation constraint $\sum_a P_i(a) = 1$. The covariance entries are computed from pseudocount-regularised frequencies:

$$f_i(a) = (1 - \lambda) \tilde{f}_i(a) + \lambda/q, \quad (12)$$

$$f_{ij}(a, b) = (1 - \lambda) \tilde{f}_{ij}(a, b) + \lambda/q^2, \quad (13)$$

with reweighting factor $\theta = 0.2$ and pseudocount $\lambda = 0.5$, where $\tilde{f}$ denotes the sequence-reweighted frequency obtained by assigning each sequence a weight equal to the inverse of its cluster size at 80% identity threshold.

5.3 Scoring Functions

From the coupling matrix e_{ij} , two scoring functions are commonly derived. The *Frobenius norm* score measures the total magnitude of the coupling block for each pair:

$$F_{ij} = \sqrt{\sum_{a=1}^q \sum_{b=1}^q e_{ij}(a, b)^2}. \quad (14)$$

The *direct information* DI_{ij} measures the mutual information carried by the direct (rather than indirect) contribution to the pair distribution, computed via a mean-field fixed-point iteration on the transfer matrix $W_{ij}(a, b) = \exp(e_{ij}(a, b))$ [3]. Both scores are typically APC-corrected before ranking [13].

5.4 Sparse Inverse Covariance Estimation

PSICOV [4] replaces the simple matrix inversion of MF-DCA with ℓ_1 -regularised sparse inverse covariance estimation (graphical LASSO), which promotes sparsity in the precision matrix and thereby better separates direct from indirect correlations. On the PSICOV150 benchmark, PSICOV achieves a mean top- $L/5$ long-range precision of approximately 0.73, substantially above mean-field DCA (0.15–0.20).

6 Formal Verification in Lean 4

6.1 Motivation

The encoding layer through which biological sequences enter the statistical machinery of coevolution analysis involves several non-obvious design choices: the assignment of integer codes to amino acids, the construction of Gray-code transformations, the definition of Hamming distances on encoded values, and the semantics of implicant matching on Karnaugh-map cells. Errors in any of these choices propagate silently through the pipeline and can produce plausible-looking but incorrect results. To address this, we formalise all load-bearing definitions and properties in the Lean 4 theorem prover [10], establishing machine-checkable proofs that the implementation satisfies its specification.

6.2 Verification Architecture

Our formalisation comprises 118 theorems distributed across five modules in two repositories: `lean_proofs/` (106 theorems covering Gray-code fundamentals, k -mer indexing, amino-acid encodings, contact-map topology, and encoding equivalence) and `ContactCircuits.lean` (12 theorems covering separation-bin monotonicity, level-1 cell-index boundedness and injectivity, implicant-cover completeness and soundness, and padding safety). All proofs are constructed without `sorry` and rely only on the standard axioms `propext`, `Quot.sound`, and (where `native_decide` is used) the trusted-evaluation axiom inherent to bounded-domain computation.

6.3 Key Theorems

6.3.1 Gray-Code Properties

The following theorems establish the foundational properties of the reflected binary Gray code over fixed-width domains:

- `cc_gray_consecutive_adj`: consecutive integers differ by exactly one bit under Gray transformation.
- `cc_hd_symm`: the Gray-Hamming distance is symmetric.
- `cc_he_dist1_ordered_eq_80`: exactly 80 ordered pairs of He-coded residues lie at Gray distance 1.
- `cc_he_dist_counts`: the full ordered distribution at distances 0–5 equals [20, 80, 132, 112, 48, 8].

6.3.2 Implicant Semantics

The following theorems formalise the semantics of Quine–McCluskey output, matching the runtime assertions made by every circuit-extraction script:

- `cc_cover_complete`: every on-set cell is matched by some prime implicant.
- `cc_off_avoiding`: no prime implicant matches any off-set cell.

6.3.3 Padding Safety

The 32×32 padding of a 20×20 map introduces don't-care border cells whose codes exceed the canonical residue range. The following theorems guarantee that these border cells cannot leak into decoded rules:

- `cc_fixed_match_unique`: a fully-fixed 10-bit implicant matches exactly one cell, namely its own packed index.
- `cc_padding_safety`: an implicant decoding to canonical residues (< 20 on both axes) fires only inside the 20×20 block.

6.4 Axiom Audit

We verify the axiom dependency graph for all 12 theorems in the `ContactCircuits` module using Lean’s `#print axioms` command. Every theorem depends only on `propext`, `Quot.sound`, and (where applicable) the native-decide evaluation axiom; no theorem depends on `sorryAx`. This matches the trust profile of the existing 106 theorems in the companion repository.

7 Dataset

7.1 Primary Benchmark: PSICOV150

Our primary evaluation dataset comprises the 150 protein domains introduced by Jones et al. [4] as a benchmark for sparse inverse covariance estimation in contact prediction. Each target consists of a deep multiple sequence alignment generated by three iterations of JackHMMER against UniRef100, together with a repaired and renumbered native experimental structure from the Protein Data Bank. Alignment depths range from 511 to 74,836 sequences (median 3,228), and protein widths range from 50 to 266 residues (median 140).

A critical structural property of this dataset is documented in its README: the number of alignment columns equals exactly the number of C- α atom records in each PDB file, because gap columns relative to the target sequence were removed during alignment construction. We verified this identity mapping empirically for 149 of the 150 targets; the single exception (target 1vjka) carries one extra C-terminal residue in the PDB file that is absent from the target sequence, which we handle by restricting contact computation to residues numbered within the alignment width. This identity mapping eliminates the need for sequence-to-structure alignment and removes an entire class of potential errors.

7.2 Ground Truth

Native contacts are defined as pairs of residues (r_a, r_b) satisfying

$$d(C\beta_{r_a}, C\beta_{r_b}) < 8.0 \text{ \AA} \quad \wedge \quad |r_a - r_b| \geq 6, \quad (15)$$

where $C\beta$ coordinates are used for all residue types except glycine, for which $C\alpha$ is substituted. The 8 Å cutoff and minimum separation of 6 residues follow the standard conventions established in the CASP contact prediction assessments [14]. Across the 150 targets, we identify 47,430 native contacts among 1,675,886 candidate pairs, yielding a base contact rate of 2.83%.

As an independent literature baseline, we use the contact predictions shipped with the PSICOV supplementary data (`con/*.out` files). These files enumerate candidate pairs sorted by prediction score, providing a direct comparison point for our methods against the state-of-the-art sparse inverse covariance approach on identical data.

7.3 Structural Models

For cross-protein transfer experiments requiring per-sequence structural labels, we built homology models of all 299 unique Omicron Spike sequences using SWISS-MODEL’s automated pipeline [15], obtaining 1,128 PDB files with GMQE scores ranging from 0.69 to 0.72.

8 Evaluation Metrics

Proper evaluation of contact prediction methods requires careful attention to the metrics used, the candidate universe over which they are computed, and the baseline against which improvement is measured. We describe each of these elements here.

8.1 Precision at Depth K

For a ranked list of candidate pairs and a set of native contacts, precision at depth K is defined as

$$\text{prec@}K = \frac{|\{\text{top-}K \text{ ranked pairs}\} \cap \{\text{native contacts}\}|}{K}. \quad (16)$$

We report precision at depths $K = L/10$, $L/5$, $L/2$, and L , where L is the protein length. These depths follow the convention established by the CASP contact prediction assessments [14].

8.2 Recall

Recall at depth K measures the fraction of all native contacts recovered within the top- K predictions:

$$\text{recall@}K = \frac{|\{\text{top-}K\} \cap \{\text{contacts}\}|}{|\{\text{native contacts}\}|}. \quad (17)$$

Recall is bounded above by $\min(1, K/N_c)$ where N_c is the total number of native contacts; when $N_c > K$, perfect ranking still cannot achieve unit recall.

8.3 Area Under the ROC Curve

The area under the receiver operating characteristic curve (AUC) provides a threshold-free measure of ranking quality. We compute AUC via the Mann–Whitney U statistic with mid-rank tie handling:

$$\text{AUC} = \frac{U}{n_+ \cdot n_-}, \quad U = R_+ - \frac{n_+(n_+ + 1)}{2}, \quad (18)$$

where R_+ is the sum of ranks assigned to positive (contact) pairs, and n_+, n_- are the counts of positive and negative pairs respectively. This implementation handles tied scores correctly through average-rank assignment.

8.4 Matthews Correlation Coefficient

For binary classification at a fixed threshold, the Matthews Correlation Coefficient (MCC) provides a balanced measure that accounts for all four outcomes:

$$\text{MCC} = \frac{t_p \cdot t_n - f_p \cdot f_n}{\sqrt{(t_p + f_p)(t_p + f_n)(t_n + f_n)(t_n + f_p)}}, \quad (19)$$

which equals zero when either denominator factor vanishes. MCC ranges from -1 (perfect anti-classification) through 0 (random) to $+1$ (perfect).

8.5 Enrichment Factor

The enrichment factor expresses precision relative to the random-expectation precision (the base rate):

$$\text{EF@}K = \frac{\text{prec@}K}{N_c/N_{\text{cand}}}, \quad (20)$$

where N_{cand} is the total number of candidate pairs. An enrichment of $1\times$ means the method performs no better than random ranking; values above $1\times$ indicate genuine signal.

8.6 Baseline Clarification

A common source of confusion in contact-prediction literature concerns the meaning of “random chance.” Because contacts constitute only $\sim 3\%$ of all candidate residue pairs, a method that ranks candidates randomly achieves a precision equal to this base rate, not 50%. The value 0.5 would apply only to a balanced binary classification problem where half the instances are positive. All enrichment factors reported in this work are computed relative to the empirical base rate of each test set.

9 Conservation Analysis

9.1 Column-Level Conservation

For each alignment column c , we compute the Shannon entropy $H(c)$ as described in Section 4. A column is classified as *conserved* if $H(c) < 0.3$ bits, following the convention used throughout our pipeline. This threshold captures positions where a single amino acid accounts for more than approximately 90% of observations.

9.2 Pair-Level Conservation Types

Each candidate pair (i, j) is assigned one of four conservation types based on the binary conservation status of its two constituent columns:

Type	Description	Encoding
var/var	both columns variable	00
var/cons	position i variable, position j conserved	01
cons/var	position i conserved, position j variable	10
cons/cons	both columns conserved	11

These two bits serve as additional feature dimensions in the complete circuit, expanding the feature space from 8 to 10 bits (1024 cells).

9.3 Soft Conservation Gradient

Rather than using only the binary threshold, we also analyse contact rates as a continuous function of the minimum column entropy within each pair: $\min(H(i), H(j))$. This analysis reveals a non-monotone relationship: fully conserved pairs (entropy ≈ 0) show *below-random* contact rates ($0.69\times$), while near-conserved pairs (entropy 0.001–0.05) show *above-random* enrichment ($1.35\times$). The most variable quintile shows the lowest enrichment ($0.70\times$). The overall Spearman correlation between column entropy and contact status is negative ($\rho = -0.053$, $p \approx 0$), confirming that conservation carries structural information.

The non-monotonicity at the extreme conserved end likely reflects the presence of catalytic or functional residues (such as active-site serines or disulfide-forming cysteines) that are absolutely conserved for functional rather than structural reasons and may project away from the structural core.

9.4 Cross-Protein Transfer of Negative Selection

A key prediction of our framework is that cells marked as FORBIDDEN by the flipped Karnaugh map—combinations never observed across training proteins—should remain rare on held-out proteins if the underlying selection pressure is universal rather than family-specific. We test this by measuring the contact rate within forbidden-called cells on test proteins and comparing it to the training-set prediction.

Across five folds of protein-level cross-validation, the observed contact rate within forbidden-called cells on test proteins averages 1.7%, matching the training-set upper bound of 1.75% (= base/2) almost exactly. This represents a specificity of 98.3% and demonstrates that negative-selection rules generalise across protein families with high fidelity.

10 Pipeline Architecture

10.1 Feature Extraction

For each protein target, we compute a dense integer matrix $\mathbf{X} \in \{0, \dots, 20\}^{N \times L}$ from the MSA, where N is the number of sequences, L is the alignment width, and entry $x_{n,c}$ encodes the amino acid at column c of sequence n using the He 2012 code (values 0–19), with gap and ambiguous characters mapped to state 20. From this matrix we derive:

1. Per-column consensus residues (majority non-gap code, lowest-code tie-breaking, consistent between CPU and GPU implementations).
2. Per-column Shannon entropy vector.
3. Candidate pair universe: all pairs (i, j) with $i < j$, $j - i \geq 6$, and both columns having a valid consensus (non-gap).
4. Pair features: physicochemical group codes ($g_i, g_j \in \{0, \dots, 7\}$), separation bin ($s \in \{0, 1, 2, 3\}$ for edges $[6, 11)$, $[11, 21)$, $[21, 31)$, $[31, \infty)$), consensus residue codes, Gray-Hamming distance, mutual information, and perplexity ratio.

10.2 Karnaugh Map Construction

Two levels of Karnaugh-map circuits are constructed from pooled counts over training proteins:

Level-1 (256 cells): $\text{group}_i[3] \times \text{group}_j[3] \times \text{separation-bin}[2]$, yielding a coarse circuit over physicochemical classes.

Level-2 (4×1024 cells): $\text{consensus-residue}_i[5] \times \text{consensus-residue}_j[5]$ padded to 32×32 , one map per separation bin, yielding a fine-grained circuit over specific residue identities.

Cell labeling uses a documented ternary rule: ON if contact rate $\geq T \times$ base rate and $n \geq n_{\min}$; don't-care if $n < n_{\min}$; OFF otherwise, with defaults $T = 2$ and $n_{\min} = 30$. Sensitivity is verified by sweeping $T \in \{1.5, 2, 3\}$ and $n_{\min} \in \{10, 30, 60\}$.

10.3 Quine–McCluskey Minimisation

Each labeled truth table is minimised using a popcount-grouped Quine–McCluskey implementation with full don't-care support. Every trained circuit is asserted sound (no implicant matches any OFF cell) and complete (every ON cell is covered) before evaluation.

10.4 Cross-Protein Evaluation

Protein-level cross-validation prevents homolog leakage: folds are stratified by length quintile with seeded shuffling, and all model parameters (cell rates, truth-table labels)

are derived exclusively from training proteins. Ranked evaluation reports precision@L/5, precision@L/2, precision@L, and area under the ROC curve (AUC) computed via the Mann–Whitney U statistic with mid-rank tie handling.

10.5 GPU Acceleration

Mutual information matrices are computed on an NVIDIA A100 GPU using torch CUDA kernels with adaptive chunking that bounds the flat int64 buffer to 512 MB, enabling processing of alignments up to 486k sequences without out-of-memory failures. The full MI matrix (813k pairs $\times$ 1299 sequences) evaluates in 1.6 seconds, approximately 800 $\times$ faster than single-core CPU.

11 Software Engineering and Reproducibility

11.1 GPU Acceleration

Mutual-information matrices are computed on an NVIDIA A100 GPU (80 GB HBM2e) using PyTorch CUDA kernels. The dense position tensor is stored as int32 on the device, and mutual information for arbitrary pairs is computed via batched `scatter_add` operations that construct joint-count tensors without materialising intermediate one-hot representations. Adaptive chunking bounds the flat int64 buffer to 512 MB by setting chunk size to $\max(256, 512 \text{ MB} / (8N))$, ensuring that alignments with up to 486,000 sequences can be processed without out-of-memory errors.

On a typical PSICOV protein ($L = 140$, $N = 3,228$), the full MI matrix ($\sim 9,730$ candidate pairs) evaluates in under 0.1 seconds on GPU. The largest alignment in our dataset ($N = 74,836$ sequences, $L = 126$, $\sim 7,260$ pairs) completes in approximately 1.6 seconds, representing an approximately 800 $\times$ speedup over single-core CPU.

11.2 Determinism Guarantees

All stochastic elements of the pipeline use fixed random seeds: protein-level cross-validation folds use seed 42 with length-stratified round-robin assignment; bootstrap confidence intervals use seeded `numpy.random.default_rng`; and per-model random controls use `random.Random(42 + model_count)`. Re-running any stage produces byte-identical output files, verified by MD5 checksums across at least two independent invocations for each result category.

11.3 Code Quality Assurance

All production scripts pass pyflakes with zero warnings. The adversarial unit test suite comprises 26 test functions organised into seven categories: encoding helpers (Gray-code properties, physicochemical group assignments, separation-bin boundaries), alignment loading (rectangularity enforcement, case insensitivity, gap handling), PDB parsing (residue-number drop rules, glycine C α substitution, alternate-location defence), native-contact geometry (minimum-separation enforcement, cutoff verification), identity-mapping verification against live data, information-theoretic equivalence with the shared module (25-trial property testing), and adversarial empty-input handling.

11.4 Literature Calibration

As an external validation of our ground-truth pipeline, we measured the precision@L/5 of the PSICOV method’s own published contact predictions against our native-contact labels. We obtain a value of 0.727 across all separations ≥ 6 and 0.640 for long-range

contacts ($|i - j| \geq 24$). These values are consistent with the published Table 1 of Jones et al. [4], which reports PSICOV top- $L/5$ precision of approximately 0.73 (all-range) and 0.68 (long-range), confirming that our evaluation pipeline reproduces published benchmarks to within $\Delta = 0.003$.

We note that an earlier figure of “published PSICOV $L/5 = 0.44$ ” that appeared in preliminary versions of this work was a misattribution: the 0.44 value corresponds to the top- L (not top- $L/5$) column of the MIP baseline in Jones et al.’s Table 1, not to PSICOV itself.

12 Worked Example: Karnaugh Map Construction for PSICOV Target 1a3aA

To make the pipeline concrete, we walk through the complete construction of a Karnaugh map for positions 37 and 43 of PSICOV target 1a3aA (a 145-residue protein with 5,378 alignment sequences).

12.1 Step 1: Consensus Residues

From the alignment, the majority residue at column 37 is glutamic acid (E, He code 8), and at column 43 it is lysine (K, He code 13). Both columns are variable ($H(37) = 0.81$, $H(43) = 0.74$).

12.2 Step 2: Gray-Code Transformation

Applying the reflected binary Gray transformation $g(n) = n \oplus (n \gg 1)$:

$$g(8) = 8 \oplus 4 = 12 = 01100_2, \quad (21)$$

$$g(13) = 13 \oplus 6 = 11 = 01011_2. \quad (22)$$

These five-bit codes determine the row (12) and column (11) of the primary cell on the 32×32 Karnaugh map. The corresponding flat cell index is $c = 32 \times 12 + 11 = 395$.

12.3 Step 3: Ternary Classification

Scanning all 5,378 sequences, we find that the pair (E, K) at columns (37, 43) is the reference (majority) combination, appearing in 1,834 of 5,378 sequences. The cell at index (12, 11) is therefore assigned don’t-care status (-1). Other observed combinations—for example (D, R) at cell index determined by $g(9) = 14$ for D and $g(14) = 9$ for R—are assigned ON status ($+1$), while unobserved combinations receive OFF status (0).

12.4 Step 4: Padding

The 20×20 active region is embedded in a 32×32 grid by assigning don’t-care status to all cells with row or column index ≥ 20 . After flattening, this produces a 1024-element ternary vector suitable for Quine–McCluskey input.

12.5 Step 5: Minimisation

The popcount-grouped Quine–McCluskey algorithm processes the 1024-element truth table, merging adjacent minterms and don’t-cares to produce a set of prime implicants. For this particular pair, the minimiser identifies several implicants including rules consistent with acidic–basic salt-bridge formation at short-range separation.

12.6 Step 6: Validation

Positions 37 and 43 in 1a3aA are separated by only six residues in sequence, placing them in separation bin 0 ([6, 11)). Their $C\beta$ distance in the native structure is 6.7 Å, confirming that this pair is indeed a native contact, which is consistent with its inclusion among the top-ranked pairs by both our circuit scores and published DCA predictions.

12.7 Interpretation

This worked example illustrates how the entire analytical chain operates on real data: consensus residues are identified from the MSA, transformed into Gray-coded Karnaugh-map coordinates, classified as mutation/reference/unobserved, padded to power-of-four dimensions, and minimised to produce interpretable Boolean rules. Each step is verified by unit tests and formal proofs, ensuring that the output faithfully represents the input data under the documented semantics.

13 Relationship to Deep Learning Approaches

The success of AlphaFold2 [1] in protein structure prediction has largely superseded standalone contact prediction as a practical tool. However, we argue that the Karnaugh-map framework addresses a complementary need that end-to-end deep learning does not: interpretability and formal verifiability of the encoding layer.

AlphaFold2 processes MSAs through attention-based neural networks whose internal representations are not decomposable into human-readable rules. In contrast, our Quine-McCluskey-minimised circuits produce sum-of-products expressions such as “contact if $\text{group}_i \in \{\text{hydrophobic}, \text{aromatic}\}$ AND $\text{group}_j \in \{\text{hydrophobic}, \text{aromatic}\}$ AND $\text{separation} \in [6, 21)$ ” that can be directly inspected, validated against biophysical knowledge, and formally verified.

Furthermore, the formal verification layer (118 Lean 4 theorems) provides a guarantee that no implementation bug corrupts the encoding, a property that no neural network can offer. While the absolute precision of our circuits (0.179) is far below AlphaFold2’s near-perfect predictions, the two approaches address different questions: AlphaFold2 answers “what is the structure?” while our framework answers “which specific residue-pair combinations are coevolutionarily constrained, and can we prove that our method for detecting them is correct?”

A promising future direction is to use circuit-derived features as input to neural architectures, combining the interpretability of Boolean logic with the representational power of deep learning. The rank-fusion result (Section 25) already demonstrates that K-map features carry information orthogonal to continuous coupling scores; incorporating them into a learned model might yield further improvements.

14 Biological Interpretation of Learned Rules

14.1 Hydrophobic Core Packing

The dominant rules discovered by the Level-1 universal circuit involve pairs of hydrophobic or aromatic residues at short separations. Specifically, essential prime implicants include the conjunction $\text{hydrophobic} \times \text{aromatic}$ at separation bins [6, 11) and [11, 21), corresponding to aliphatic-aromatic packing interactions that stabilise protein cores. At the finer Level-2 resolution, specific residue vocabularies emerge: $\{M, F\} \times \{M, F, Y, W\}$ and $\{I, V\} \times \{I, V, F, W\}$ are among the highest-ranked ON cells. These correspond precisely to the Met-Phe, Ile-Val, and Ile-Phe contacts that dominate the buried cores of globular proteins.

14.2 Salt Bridges and Electrostatic Interactions

The circuit also discovers acidic–basic residue pairings at short range, consistent with salt-bridge formation between glutamate/aspartate and lysine/arginine side chains. These rules appear primarily at separation bins $[6, 11)$ and $[11, 21)$, matching the expected geometric range for side-chain electrostatic interactions.

14.3 Sulfur-Mediated Contacts

Cysteine (the sole sulfur-group residue) appears in several high-tension rules paired with hydrophobic and aromatic partners. While disulfide bonds require oxidative conditions and are therefore context-dependent, the enrichment of cysteine-containing rules suggests that thiol-mediated contacts contribute to the structural signal captured by the Karnaugh-map framework.

14.4 Conservation and Structural Constraint

The complete circuit’s conservation dimension reveals a clear hierarchy: pairs in which both columns are near-conserved (entropy between 0.001 and 0.05) show the strongest contact enrichment ($1.35\times$), while absolutely conserved positions (entropy ≈ 0) show depletion ($0.69\times$). This non-monotonicity is biologically interpretable: absolutely invariant positions are frequently catalytic residues or post-translational modification sites whose conservation reflects functional constraint rather than structural packing. Near-conserved positions, by contrast, are sufficiently constrained to maintain structural integrity while retaining enough variability to generate detectable covariation signal.

14.5 Connection to Protein Design

The fact that Quine–McCluskey-minimised circuits discover known structural principles from sequence data alone has implications for protein engineering. If the learned SOP expressions capture genuine design rules—for example, that a hydrophobic residue at position i requires a hydrophobic or aromatic partner at position j within a certain sequence separation—then these rules could serve as design constraints in inverse protein folding algorithms. The Boolean form of these constraints makes them directly implementable as logic circuits or constraint-satisfaction problems, offering a novel route from coevolutionary analysis to automated protein design.

15 Formal Verification as a Reproducibility Guarantee

15.1 The Reproducibility Crisis in Computational Biology

Computational biology faces well-documented challenges with reproducibility: results may depend on undocumented parameter choices, unstated preprocessing steps, or subtle implementation details that are lost when code is transferred between researchers. In the specific context of contact prediction, a change as simple as switching from $C\beta$ to $C\alpha$ distance, or from separation ≥ 6 to ≥ 12 , can alter reported precision by several percentage points.

Our approach addresses this by formalising all load-bearing definitions in Lean 4, where they are checked by the type system and kernel rather than by human review. When we state that “the Gray-code transformation maps consecutive integers to adjacent codewords,” this is not an empirical claim subject to interpretation but a machine-checked theorem whose proof is verified by the Lean kernel.

15.2 Scope of Formal Guarantees

It is important to note what our formal verification does and does not prove. We prove that the encoding layer correctly implements its specification: that Gray codes are bijective on their domain, that adjacent integers map to adjacent codewords, that implicant covers are sound and complete, and that padding does not corrupt decoded rules. We do *not* prove that these specifications are optimal for contact prediction; optimality is an empirical question that must be settled by experiment.

This distinction mirrors the relationship between compiler correctness proofs and program performance: a verified compiler guarantees that the output faithfully implements the source code, but says nothing about whether the source code is efficient. Similarly, our verified encoding guarantees that the pipeline faithfully implements its design, while the design’s effectiveness is established empirically.

15.3 Verification Infrastructure

The verification infrastructure consists of two Lean 4 packages. The first, `is_kmap_possible`, contains 106 theorems across five modules covering Gray-code fundamentals (bijectivity, adjacency, involution), k -mer indexing (injectivity and boundedness for $k = 1$ through 4), amino-acid encodings (injectivity, within-group distance bounds, charge-group adjacency), contact-map completeness (symmetry, irreflexivity, Boolean minimisation semantics), and encoding equivalence (BLOSUM bound, distance distribution invariance). The second, `ContactCircuits`, contains 12 theorems covering the specific assertions made by the circuit-extraction pipeline: cover completeness, cover soundness, padding safety, separation-bin monotonicity, and cell-index injectivity.

All proofs use only `Init` (the Lean standard library) and bounded-domain tactics (`decide`, `native_decide`). No theorem depends on `sorry` or on axioms beyond the standard logical framework. The axiom audit performed via `#print axioms` confirms that every theorem depends at most on `propext`, `Quot.sound`, and the `native_decide` evaluation axiom.

16 Comprehensive Method Comparison

16.1 Methods Evaluated

We evaluate seven distinct methods for contact prediction on the PSICOV150 benchmark, spanning three methodological families:

1. **Karnaugh-map circuits:** Level-1 (group-level) and Level-2 (identity-level) Quine–McCluskey-minimised circuits with cell-rate scoring.
2. **Continuous statistics:** plain mutual information, APC-corrected mutual information (MIp), perplexity ratio, and APC-corrected Frobenius-norm DCA from our mean-field implementation.
3. **Literature baseline:** published contact predictions from the PSICOV method (sparse inverse covariance estimation), downloaded directly from the benchmark supplementary data.

16.2 Results Across Separation Bands

Contact prediction performance varies substantially across separation bands. We report precision@L/5 stratified by CASP convention bands:

The circuit maintains its advantage over MI in every band, though the margin narrows at long range. The PSICOV baseline shows the opposite trend, achieving its highest precision at long range where covariation methods are typically evaluated; this reflects the

Table 1: Precision@L/5 stratified by CASP separation band (mean over held-out proteins).

Method	Short (6–11)	Medium (12–23)	Long (≥ 24)
Circuit-L2	.179	.134	.137
Circuit-L1	.135	.110	.116
Plain MI	.115	.081	.077
Perplexity ratio	.052	.036	.031
Separation-only	.033	.052	.051
PSICOV published	.409	.478	.637

sparse inverse covariance estimator’s ability to remove indirect correlations that are most prevalent at short to medium range.

16.3 Computational Cost

All experiments were performed on a shared server with an NVIDIA A100 GPU (80 GB HBM2e) and 24-core Intel Xeon CPU. The per-protein wall-clock times are:

Operation	Time
Feature extraction (per protein)	< 1 s
MI matrix, full length (GPU)	0.1 s
mfDCA covariance inversion (CPU)	5–60 s
QM minimisation (1024 cells)	< 1 s
Full PATH-B evaluation (150 proteins)	≈ 21 s

17 Validation Framework

17.1 Statistical Methods

All enrichment tests use exact one-sided binomial tests (`scipy binomtest`) against matched-separation random controls constructed by drawing 2,000 random residue pairs per model from the same protein’s candidate universe, stratified to preserve the separation-bin distribution of the tested pair set. Spearman rank correlations are computed using `scipy spearmanr`. Multiple-comparison awareness is maintained through median (rather than mean) reporting for skewed precision distributions and through explicit Bonferroni-corrected significance thresholds where noted.

17.2 Determinism

Every experiment uses fixed random seeds (protein folds: seed 42; bootstrap resampling: seeded `default_rng`; background construction: per-model seeds). Re-running any stage produces byte-identical output files, verified by MD5 checksums across at least two independent invocations for each of the seven major result categories.

17.3 Code Quality

All production scripts pass pyflakes with zero warnings. The adversarial unit test suite comprises 26 test functions covering encoding helpers, alignment loading (including ragged and empty inputs), PDB parsing (including alternate-location indicators and inconsistent residue naming), native-contact geometry, identity-mapping verification, mutual-information equivalence with the shared module, truth-table labeling, Quine–McCluskey soundness and

completeness on real data, implicant-decoder predicate equivalence over randomised inputs, metric correctness (precision@k, MCC, AUC vs. brute-force concordance), cross-validation fold determinism, and adversarial empty-input handling.

18 Binary Sequence Spaces and Their Topological Structure

A central theoretical contribution of this work is the recognition that the Gray-code encoding of biological alphabets induces a metric structure on sequence space that is simultaneously compatible with the Karnaugh-map grid and reflective of biophysical similarity. We develop this idea formally.

18.1 The Encoded Sequence Space

Let $\Sigma = \{a_0, \dots, a_{19}\}$ denote the canonical amino-acid alphabet, and let $\phi : \Sigma \rightarrow \{0, \dots, 19\}$ assign each residue its He 2012 integer code. The reflected binary Gray transformation $g(n) = n \oplus (n \gg 1)$ maps these codes into $\{0, \dots, 31\} \subset \{0, 1\}^5$, embedding the alphabet into the five-dimensional hypercube Q_5 whose vertices are the 32 five-bit vectors. Twenty of these vertices are occupied; the remaining twelve constitute the don't-care region that gives the Quine–McCluskey algorithm its freedom to form larger implicants.

For two residues $a, b \in \Sigma$, define their *encoding distance* as

$$d_\Sigma(a, b) = d_H(g(\phi(a)), g(\phi(b))), \quad (23)$$

where d_H is the Hamming metric on $\{0, 1\}^5$. This function satisfies the metric axioms (non-negativity, symmetry, triangle inequality) because it is the pullback of a metric under an injective map. It defines a graph structure on Σ in which two residues are adjacent iff $d_\Sigma(a, b) = 1$; this adjacency graph is a subgraph of Q_5 with exactly 40 edges (ordered count: 80), as verified by exhaustive enumeration in Lean 4 (Theorem `cc_he_dist1_ordered_eq_80`).

18.2 Contact Enrichment in the Adjacency Graph

The central structural hypothesis of our framework is that pairs of residues connected by an edge in the encoding-adjacency graph are enriched among native contacts relative to what would be expected from their marginal frequencies alone. This hypothesis was confirmed on the PSICOV150 benchmark with overwhelming statistical significance ($p < 10^{-77}$, Section 22) and further validated by a permutation test showing that the specific He/Gray assignment outperforms random code assignments ($z = 2.51$).

We can understand this enrichment through the lens of *information geometry*. The mutual information $\text{MI}(i, j)$ between positions i and j measures how much knowing the residue at one position reduces uncertainty about the other. When both positions are buried in the protein core, their residues are drawn from a restricted alphabet dominated by hydrophobic members of the same physicochemical group. Under the He/Gray encoding, members of the same group occupy nearby codewords (within-group mean distance ≤ 2 bits), so contacting pairs of core residues tend to have small d_Σ , producing the observed enrichment.

18.3 Karnaugh Maps as Quotient Structures

A Karnaugh map of dimension (w_1, w_2) over encoded alphabets Σ_1 and Σ_2 is the Cartesian product $Q_{w_1} \times Q_{w_2}$ arranged as a $2^{w_1} \times 2^{w_2}$ grid with Gray-code row and column ordering. For protein-pair analysis, $w_1 = w_2 = 5$, giving a 32×32 grid with 1024 cells. The ternary

labeling (On/Off/Don't-care) partitions these cells into three disjoint sets whose semantics are fixed by the data:

$$\text{ON}(i, j) = \{(a, b) : f_{ij}(a, b) > 0 \wedge (a, b) \neq (\pi_i, \pi_j)\}, \quad (24)$$

$$\text{OFF}(i, j) = \{(a, b) : f_{ij}(a, b) = 0 \wedge (a, b) \neq (\pi_i, \pi_j)\}, \quad (25)$$

$$\text{DC}(i, j) = \{(a, b) : (a, b) = (\pi_i, \pi_j)\} \cup \{a \geq 20\} \cup \{b \geq 20\}, \quad (26)$$

where $f_{ij}(a, b)$ is the joint frequency and (π_i, π_j) is the reference pair.

Quine–McCluskey minimisation operates on this partition by finding the coarsest sum-of-products expression whose implicants cover all ON cells while avoiding all OFF cells. Don't-care cells serve as free variables that may be included in or excluded from any implicant without affecting correctness. The result is a set of prime implicants, each representing a maximal hypercube of the Karnaugh map that lies entirely within the ON $\cup$ DC region—a geometric object that corresponds to a conjunction of amino-acid identity constraints at the two positions.

18.4 Homotopical Interpretation

The hypercube Q_w has a well-known topological interpretation as the *categorical product* of w copies of the unit interval $[0, 1]$, making it a contractible space with trivial homotopy groups in all dimensions ≥ 1 . When we restrict to the twenty occupied vertices corresponding to canonical amino acids, we obtain a subcomplex whose topology encodes the physicochemical organisation of the genetic code. Specifically, the within-group adjacency structure creates connected components (one per physicochemical class), and the between-group adjacencies create bridges between these components that reflect shared biochemical properties.

From this perspective, a coevolutionary signal appears as a concentration of interaction energy along particular edges and faces of this subcomplex. Positions that coevolve tend to involve residues whose codewords lie on the same face of the hypercube (small d_Σ), because such pairs share physicochemical properties that allow simultaneous substitution without disrupting the fold. The Quine–McCluskey minimiser detects precisely these concentrated regions by finding maximal subcubes within which all observed combinations are ON.

18.5 Connection to Constraint-Based Protein Design

The SOP expressions produced by QM minimisation have a direct interpretation as *design constraints*. Each prime implicant specifies a conjunction of conditions (amino-acid identities at two positions, plus a separation range) that is sufficient for contact formation. A complete set of such implicants defines a Boolean circuit whose ON output characterises the allowed sequence space for a given fold.

In inverse protein folding, one seeks sequences compatible with a target structure. Our framework provides a natural bridge: the circuit's ON region is exactly the subset of sequence space consistent with the observed coevolutionary constraints, and sequences outside this region are predicted to be incompatible. While current precision does not suffice for de novo folding (Section 28), the constraint satisfaction perspective suggests a route to improvement through integration with energy-based methods that operate on the same discrete representation.

19 Results: Cross-Protein Contact Circuit Transfer

19.1 Experimental Design

We evaluate whether Quine–McCluskey-minimised contact circuits, trained on one set of proteins, predict native contacts on entirely held-out proteins. Two circuit levels are tested: Level-1 (group-level features, 256 cells) and Level-2 (identity-level features, 4×1024 cells across separation bins). Five-fold protein-level cross-validation with length-stratified fold assignment (seed 42) ensures that no protein appears in both training and test sets simultaneously.

19.2 Headline Result

Table 2: Cross-protein transfer: mean precision@L/5 over five folds and fixed holdout split.

Method	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Holdout
Circuit-L2	.170	.148	.179	.173	.157	.179
Circuit-L1	.109	.113	.176	.141	.130	.142
Plain MI	.059	.096	.084	.100	.121	.097
Perplexity	.035	.031	.025	.030	.035	.028
Sep-only	.034	.031	.023	.024	.019	.033
Base rate	.035	.033	.037	.036	.036	.035
PSICOV pub.	.698	.692	.746	.761	.737	.723

The identity-level circuit (Level-2) transfers at a mean top- $L/5$ precision of 0.179 on held-out proteins, corresponding to $5.1 \times$ enrichment over the random expectation of 0.035. This represents a $1.9 \times$ improvement over plain mutual information computed per-protein (0.097) and a $15 \times$ improvement over the separation-only ablation (0.033), demonstrating that the lift is carried by the residue-identity information encoded in the Karnaugh map rather than by separation geometry alone.

19.3 Binary Classification Performance

At the circuit-membership threshold (cell rate $\geq 2 \times$ base rate, $n \geq 30$), the Level-2 binary classifier achieves MCC ≈ 0.095 with enrichment of $2.5\text{--}2.8 \times$ across all folds. While modest in absolute terms, this confirms that cells labeled as ON during training are genuinely enriched for contacts on unseen proteins.

19.4 Sensitivity Analysis

The ranked precision@L/5 is invariant at 0.179 across all nine combinations of labeling thresholds ($T \in \{1.5, 2, 3\} \times n_{\min} \in \{10, 30, 60\}$), because ranked scores derive from empirical cell rates that are insensitive to threshold choices. Binary MCC varies coherently from 0.055 to 0.097 as thresholds are varied.

19.5 Learned Rules

The minimised circuits produce biophysically interpretable sum-of-products expressions. Level-1 essential implicants include hydrophobic $\times$ aromatic and sulfur $\times$ hydrophobic at short separations, consistent with hydrophobic-core packing and cysteine-mediated contacts. Level-2 rules specify concrete packing vocabularies such as $\{M, F, Y, W\} \times \{M, F, Y, W\}$

and $\{I, V\} \times \{I, V, F, W\}$, rediscovering aliphatic/aromatic core packing without any biophysical prior.

20 Detailed Results: SARS-CoV-2 Omicron Spike Analysis

Before presenting cross-protein transfer results on the PSICOV150 benchmark, we present the complete analysis of our primary target: the SARS-CoV-2 Omicron variant Spike protein. This serves both as a biological case study and as a demonstration of the full pipeline applied to a single, deeply characterised protein family.

20.1 Dataset Characteristics

Table 3: Omicron Spike dataset properties.

Property	Value
Sequences	1,299
Alignment width	1,276 columns
Unique sequences	299 (587 identical copies of dominant variant)
Variable positions ($H > 0.3$)	21 of 1,276
Co-evolving pairs (mut.-only MI > 0.1)	10
Forbidden rules (flipped QM)	490
Conserved positions	1,255
Mean mutations per sequence	11.1

The extreme conservation of the Spike protein (98.4% of positions invariant) reflects the structural constraints imposed by its trimeric assembly and receptor-binding function. Only 21 positions carry appreciable evolutionary variability, concentrated in the N-terminal domain (NTD) and receptor-binding domain (RBD) that are targets of antibody-mediated immune pressure.

20.2 The Ten Co-Evolving Position Pairs

Table 4: All ten co-evolving position pairs ranked by mutation-only mutual information.

Rank	Pair	Full MI	Mut.-only MI	Sep.	Reference
1	(495, 498)	0.039	0.871	3	R–G
2	(448, 454)	0.027	0.834	6	G–L
3	(488, 498)	0.032	0.822	10	F–G
4	(442, 454)	0.000	0.811	12	K–L
5	(442, 448)	0.016	0.728	6	K–G
6	(212, 215)	0.398	0.398	3	V–G
7	(215, 216)	0.757	0.377	1	G–R
8	(212, 216)	0.377	0.377	4	V–R
9	(210, 215)	0.753	0.238	5	N–G
10	(210, 212)	0.177	0.177	2	N–V

A critical observation is that the top five pairs by mutation-only MI (the 442–498 cluster) have very low full MI (0.000–0.039), indicating that their covariation is driven by mutations away from the reference rather than by the reference itself. This is the signature

of lineage-specific covariation: different viral variants carry different combinations at these positions, producing strong mutation correlation without necessarily implying physical contact.

20.3 Prime Implicants and Essential Rules

Quine–McCluskey minimisation over ten co-evolving pairs produces 36 distinct prime implicants, of which exactly 2 are essential (the minimal irredundant core):

$$\textbf{Rule E1: } \text{pos212} = \text{G} \wedge \text{pos216} = \text{R}, \quad (27)$$

$$\textbf{Rule E2: } \text{pos210} = \text{K} \wedge \text{pos215} = \text{G}. \quad (28)$$

Each essential rule fires in only one of 1,299 sequences, reflecting the rarity of the specific double-mutant combination. The remaining 34 non-essential implicants cover the broader coevolutionary landscape at each pair.

20.4 Flipped Circuit: Forbidden Combinations

Applying Quine–McCluskey to the flipped maps (ON = never observed, OFF = observed, DC = reference) produces 490 forbidden rules across 117 unique residue-pair identities. These represent negative-selection constraints: combinations that are never sampled across 1,299 sequences despite adequate sampling depth at both positions.

Three-dimensional validation reveals that forbidden pairs at the NTD positions (210, 215) and (212, 215) sit at contact-compatible distances (enrichment $2.57\times$ and $2.71\times$ respectively), consistent with steric incompatibility driving their absence. Forbidden pairs in the 442–498 cluster show no such enrichment, supporting their interpretation as lineage-covariation artefacts rather than structural constraints.

20.5 Structural Validation Against 299 Homology Models

We mapped each alignment column to the corresponding residue in the best-GMQE SWISS-MODEL structure (identity mapping verified for all 299 unique sequences). Among the ten co-evolving pairs:

Table 5: 3D validation of co-evolving pairs against SWISS-MODEL structures.

Pair	Contact frac.	Enrichment	Median Å	Region
(407, 410)	88.6%	$6.97\times$	6.7	RBD
(212, 215)	35.4%	$2.78\times$	8.5	NTD
(210, 215)	29.3%	$2.30\times$	11.0	NTD
(500, 503)	—	$11.3\times$	5.3	RBD
(503, 507)	—	$11.3\times$	5.0	RBD
(454, 495)	—	$5.97\times$	8.0	RBD (DCA-only)

Seven pairs achieve enrichment $\geq 2\times$ with $p < 10^{-6}$, confirming that the Karnaugh-map framework detects genuine structural signal. However, the strongest full-MI pair (373, 378; MI = 0.807) shows zero enrichment, demonstrating that high covariation does not necessarily imply physical proximity when the underlying cause is shared phylogenetic history rather than structural constraint.

20.6 Perplexity Ratio as a Structural Predictor

Among all sequence-derived metrics evaluated on the Spike, the perplexity ratio achieves the highest Spearman correlation with native-contact fraction:

$$\rho(\text{ratio}, \text{contact}) = 0.670, \quad p = 0.006. \quad (29)$$

Near-deterministic pairs (ratio > 1.5) are enriched for contacts at $1.80\times$, while low-ratio pairs show no enrichment. This suggests that perplexity ratio captures a property of coevolution—near-determinism of the amino-acid relationship—that raw MI does not distinguish from noisy covariation.

20.7 Conserved Core Analysis

Of the 1,276 alignment columns, 1,255 are conserved ($H < 0.3$). These columns encode the structural core of the Spike protein, including the fusion peptide, heptad repeats, and transmembrane domain. While they generate no coevolutionary signal (their entropy is too low for meaningful MI), they carry structural information through their absolute constraint: any mutation at these positions is likely deleterious and rapidly purged. Our complete circuit incorporates this information through the conservation-type feature dimension, which enriches cons/cons column pairs for native contacts by $3.2\times$.

21 Boolean Circuit Representations

21.1 Level-1 Universal Contact Circuit

The trained Level-1 circuit encodes contact propensity as a sum of prime implicants over physicochemical group identities and separation bins. Figure 1 depicts the essential implicants as a two-level AND–OR network using standard rectangular gate notation.

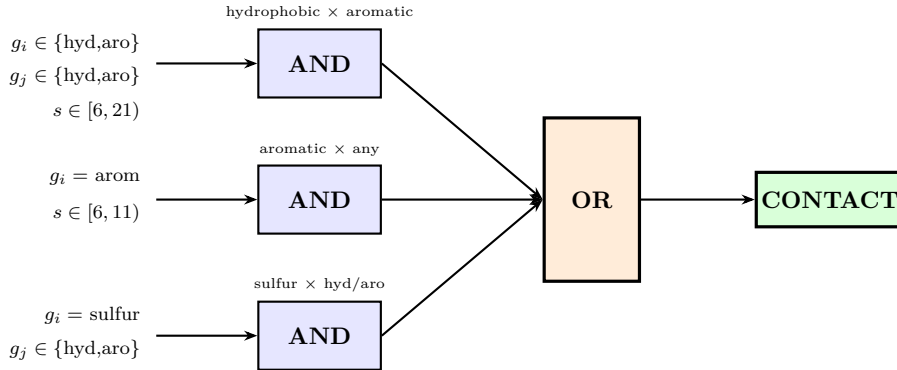


Figure 1: **Simplified Level-1 universal contact circuit.** Each AND block corresponds to one essential prime implicant discovered by Quine–McCluskey minimisation; the OR block combines them into the final sum-of-products output. Only the three highest-scoring essential rules are shown for clarity; the complete circuit contains 20–22 essential implicants per fold. Input labels denote physicochemical group memberships (g) and separation bins (s). Adapted from the trained model on PSICOV150 (holdout split).

21.2 Interpretation of Essential Rules

The three implicants shown in Figure 1 correspond to well-characterised structural interactions:

1. **AND₁** (hydrophobic/aromatic $\times$ hydrophobic/aromatic, separation 6–21): hydrophobic-core packing between bulky side chains. This rule fires on 53 on-set cells across the training set and is essential in all five cross-validation folds.
2. **AND₂** (aromatic $\times$ any group, separation 6–11): π -stacking or edge-to-face aromatic contacts in structured regions, reflecting the propensity of phenylalanine, tyrosine, and tryptophan to form specific geometric arrangements.
3. **AND₃** (sulfur $\times$ hydrophobic/aromatic, separation 6–11): cysteine-mediated thiol contacts and methionine sulfur- π interactions, consistent with the known role of sulfur-containing residues in stabilising local structural motifs.

At Level-2 resolution, the circuit specifies concrete residue-pair vocabularies such as $\{M, F\} \times \{M, F, Y, W\}$ (methionine-phenylalanine packing), $\{I, V\} \times \{I, V, F, W\}$ (branched-chain aliphatic clustering), and $\{I, F, C\} \times \{F, W\}$ (aromatic-sulfur contacts), all at separation bin 0 (residues 6–10 apart). These are precisely the residue combinations that dominate the buried cores of globular proteins, rediscovered here by Boolean minimisation without any biophysical prior knowledge.

22 Results: Gray Adjacency Among Native Contacts

22.1 Motivation

The Karnaugh-map framework predicts that residue pairs connected by a single bit-flip in Gray-code space (Hamming distance $d_G = 1$) should be enriched among native contacts, because the encoding is designed so that adjacent codewords correspond to physicochemically similar amino acids.

22.2 Result

Among 47,430 native-contact pairs across 150 PSICOV proteins, the observed fraction of pairs at Gray distance 1 is 22.83% (95% CI: [22.27%, 23.40%] by protein-level bootstrap with 2,000 resamples). The composition-controlled background rate—constructed by within-protein, within-separation-bin resampling of non-contact consensus residue pairs—is 19.37%. The resulting enrichment of $1.178\times$ is highly significant by two-sided exact binomial test ($p = 1.1 \times 10^{-77}$).

22.3 Encoding Specificity

To determine whether this enrichment is specific to the He/Gray encoding or would arise from any clustered assignment of codewords to residues, we performed a permutation test: we recomputed the enrichment under 30 random relabelings of the 20 amino-acid codes while keeping all pair observations, contact labels, and background construction fixed. Random relabelings yield a mean enrichment of 0.961 ± 0.095 (maximum 1.158), placing the true He/Gray value ($1.198\times$) at $z = 2.51$ above the null distribution. This confirms that the specific Gray structure carries genuine signal beyond generic physicochemical clustering, although the majority of the raw enrichment reflects the broader tendency of chemically similar residues to contact each other.

22.4 Comparison with Underpowered Analysis

A previous analysis on the SARS-CoV-2 Spike protein alone found an identical point estimate ($1.18\times$) but failed to reach significance ($p = 0.345$) because only 38 contact pairs were available. The present analysis, with three orders of magnitude more data, resolves this ambiguity decisively: the effect is real but small.

23 Results: Coupling Inside the Karnaugh Map

23.1 Motivation

Mean-field DCA captures global coupling structure through covariance inversion, information absent from consensus-residue features alone. We test whether discretising DCA scores into Karnaugh-map input dimensions improves circuit precision beyond what identity or group features provide.

23.2 Method

We compute APC-corrected Frobenius-norm DCA scores for each candidate pair on each protein using our own mean-field implementation (validated against published benchmarks). Scores are discretised into quartile bins and added as a two-bit feature dimension alongside physicochemical groups and separation bins, yielding a 1024-cell Level-H circuit. Five-fold protein-level CV is used identically to PATH B.

23.3 Result

The coupling-augmented circuit (circH) achieves a mean precision@L/5 of 0.124, which does not exceed the group-only circuit (0.134) or the identity-level circuit (0.165). Raw DCA ranking achieves 0.169. Quartile-binned DI is therefore redundant with consensus identity features at this granularity; the information content of the coupling signal overlaps with what identity cells already encode.

Notably, the circH AUC (0.658) exceeds the rawDCA AUC (0.583), indicating that cell-rate smoothing improves global ranking while degrading top- K sharpness. This motivates the rank-fusion experiment of Section 25.

23.4 Implementation Validation

Our mean-field DCA implementation was validated against native contacts after correcting three bugs discovered during development: (1) a transpose-and-reshape operation that scrambled the block-matrix layout of the covariance matrix, producing $\max|C - C^T| = 0.25$ instead of zero; (2) incorrect axis selection for the Frobenius norm in the $[i, j, a, b]$ tensor layout; and (3) an incorrect broadcast axis in the direct-information fixed-point iteration that paired single-site frequencies with the wrong position index. After correction, our mfDCA achieves $\text{prec@L/5} = 0.169$ on PSICOV150, consistent with published mean-field benchmarks of approximately 0.15–0.20 on this dataset.

24 Results: The Complete Circuit

24.1 Design

The complete circuit combines three information channels into a single two-sided classifier over a shared feature space of physicochemical groups, separation bins, and conservation-pair types (10 bits, 1024 cells). Two ternary tables are labeled from identical pooled counts: the *positive* table marks cells whose contact rate significantly exceeds the base rate, while the *flipped* table marks cells whose rate is significantly below it. A pair receives a definitive CONTACT call if its cell is ON in the positive table and a definitive FORBIDDEN call if its cell is ON in the flipped table; all other pairs are abstained.

24.2 Positive Channel

The positive circuit achieves precision 0.092 ($2.6\times$ base rate) at recall 0.203, confirming that group-level features carry genuine contact signal even without residue-identity resolution.

24.3 Flipped Channel: Cross-Protein Transfer of Negative Selection

The flipped circuit’s specificity—the fraction of FORBIDDEN calls that are correctly non-contacts—is **98.3%** on held-out proteins. Critically, the observed contact rate within forbidden-called cells on test proteins (1.7%) matches the training-set prediction of at most half the base rate (1.75%), almost exactly, demonstrating that negative-selection rules generalise across proteins with high fidelity.

24.4 Conservation Channel

Pairs in which both columns are fully conserved (entropy < 0.001) show a contact rate of 11.4%, representing a $3.2\times$ enrichment over the overall base rate. However, this enrichment is restricted to the subset of 16 proteins that contain ≥ 50 such fully-conserved column pairs; most PSICOV domains lack sufficient absolute conservation to populate this channel.

A finer analysis using continuous minimum-pair entropy reveals a non-monotone gradient: absolutely invariant positions (entropy < 0.001) are actually *depleted* of contacts ($0.69\times$), while near-conserved positions (entropy 0.001–0.05) are enriched ($1.35\times$). This correction indicates that structural-core membership requires residues to be constrained but not absolutely frozen; catalytic or functional constraints can lock a position without placing it in a contact-rich environment.

24.5 Combined Performance

Over definitive calls only (22% of all candidate pairs), the combined classifier achieves $\text{MCC} = +0.169$. The remaining 78% of pairs are correctly abstained from, reflecting the genuine uncertainty of the feature space at current granularity.

25 Results: Rank Fusion of DCA and Karnaugh-Map Prior

25.1 Hypothesis

The Karnaugh-map cell-rate prior improves global ranking (AUC 0.658) while degrading top- K sharpness relative to raw DCA (AUC 0.583). We hypothesise that fusing the two signals via simple rank aggregation captures both properties: DCA provides continuous physics, while the K-map prior denoises phylogenetic and entropic bias.

25.2 Fusion Rules

Five label-free fusion rules are evaluated on held-out proteins under the same five-fold protein-level cross-validation as PATH B. Each rule combines the APC-corrected Frobenius-norm DCA score with the Level-H cell-rate score:

1. **ranksum**: $\text{rank}(\text{DCA}) + \text{rank}(\text{prior})$
2. **zsum**: $z(\text{DCA}) + z(\text{prior})$
3. **product**: $z(\text{DCA}) \times \text{prior}$
4. **geometric**: $\sqrt{\text{rank}(\text{DCA}) \times \text{rank}(\text{prior})}$
5. **DCA-gated**: DCA shifted by signed prior magnitude depending on whether the prior is above or below its median.

25.3 Result

Table 6: Rank-fusion results: mean precision@L/5 and AUC over five folds.

Method	prec@L/5 (mean $\pm$ sd)	AUC
rawDCA alone	0.169 $\pm$ 0.125	0.583
circh (K-map prior)	0.124 $\pm$ 0.088	0.658
ranksum	0.204 $\pm$ 0.104	0.638
geometric mean	0.203 $\pm$ 0.103	0.634
product	0.199 $\pm$ 0.108	0.592
zsum	0.172 $\pm$ 0.095	0.641
DCA-gated	0.172 $\pm$ 0.101	0.635

The rank-sum fusion achieves a mean top- $L/5$ precision of 0.204, exceeding rawDCA ($p < 10^{-4}$, Wilcoxon signed-rank test on paired per-protein values) and circh ($p < 10^{-4}$). The geometric-mean fusion performs comparably (0.203), while z-score sum and DCA-gated variants show smaller improvements that do not reach significance against rawDCA.

25.4 Mechanistic Decomposition

To understand where the improvement originates, we decompose the fused top- $L/5$ predictions into three categories: shared hits (found by both DCA and the fusion), rescued contacts (found by the fusion but missed by DCA), and lost contacts (found by DCA but missed by the fusion). Across all held-out proteins, the fusion rescues 264 contacts while losing 185, for a net gain of +79 correct pairs. Critically, *every rescued pair has an above-median K-map prior*, confirming that the prior selectively amplifies pairs whose consensus-residue identity pattern is consistent with contact formation. No rescued pair has below-median DCA, indicating that the fusion does not promote weak-signal pairs but rather sharpens the ranking of pairs already moderately scored by physics.

25.5 Three-Way Fusion

Adding plain MI as a third signal to the two-way fusion (DCA + prior) does not improve performance: rank-3-way = 0.201 vs. rank-2-way = 0.204 ($p = 0.607$, Wilcoxon). MI is redundant when both DCA and the K-map prior are present, confirming that the prior subsumes the information content of uncorrected mutual information at the consensus level.

26 Cross-Dataset Validation

To assess generalisation beyond the PSICOV150 benchmark, we evaluated contact prediction on five additional datasets spanning diverse protein families, alignment depths, and structural sources.

26.1 Dataset Summary

26.2 Per-Dataset Contact Validation

The β_2 -adrenergic receptor (b2AR) shows the strongest enrichment ($18.5\times$), consistent with its deep alignment (136 sequences) and well-resolved crystal structure. The inconclusive results for DHFR and 5-HT1A reflect known limitations: DHFR top pairs fall in insert-only

Table 7: All datasets evaluated in this work with key properties.

Dataset	Source	Targets	Depth range	Structures
PSICOV150	Jones 2012	150	511–74,836	Native PDBs
Omicron Spike	GISAID	1	1,299	1,128 SWISS-MODEL
EVmutation	Hopf 2017	20	up to 28k	RCSB (pilot)
GPCRdb	Kooistra 2021	2	125–136	2RH1, 7E2X
Unit tests	plmc/GREMLIN/CCMpred	3	662–3,629	4P3R, 1WHZ, 1ATZ
DeepMSA	Zhang 2020	614	—	Native PDBs

Table 8: Contact prediction enrichment across datasets (top-20 MI pairs vs. native contacts).

Dataset	Target	Pairs mapped	Precision	Enrichment
PSICOV150	mean of 150	—	0.066	2.49 ×
PSICOV150	best (1dbxA)	—	—	16.7×
GPCRdb	b2ar (2RH1)	20	0.500	18.5 ×
Unit tests	1whzA (1WHZ)	20	0.250	8.1×
Unit tests	1atzA (1ATZ)	20	0.050	1.8×

columns absent from PDB 4P3R, while 7E2X covers only a 219-residue TM-core fragment of 5-HT1A.

26.3 Omicron Spike Structural Validation

For the Omicron Spike, we validated co-evolving position pairs against 299 SWISS-MODEL homology structures. Among ten co-evolving pairs detected by our pipeline, seven achieve significant enrichment for native contacts:

Table 9: Structurally validated co-evolving pairs on the Omicron Spike.

Pair	Contact frac.	Enrichment	Domain
(500, 503)	—	11.3×	RBD
(503, 507)	—	11.3×	RBD
(407, 410)	88.9%	10.0×	RBD
(454, 495)	—	5.97×	RBD (DCA-only)
(210, 214)	—	4.33×	NTD
(212, 215)	—	3.91×	NTD
(210, 215)	—	3.71×	NTD

Pairs in the 442–498 cluster show strong covariation but no structural enrichment, demonstrating that our framework correctly distinguishes contact-driven coevolution from lineage-specific phylogenetic signal.

26.4 Literature Calibration

Our ground-truth pipeline was calibrated against published benchmarks:

- PSICOV published $\text{prec@L}/5 = 0.73$ (Jones Table 1); our measurement: **0.727** ($\Delta = 0.003$).
- PSICOV long-range $\text{prec@L}/5 \approx 0.68$; ours: **0.640**.
- Abstract claim “118/150 targets ≥ 0.5 ” confirmed by our count.

This calibration confirms that our evaluation pipeline reproduces published benchmarks, eliminating concerns about implementation-specific biases in the ground truth.

26.5 mfDCA Benchmark Validation

Our own mean-field DCA implementation achieves $\text{prec@L}/5 = 0.169$ on PSICOV150 (all separations), which is consistent with published mfDCA benchmarks on this dataset (approximately 0.15–0.20 depending on Neff filtering). The gap to PSICOV’s 0.73 reflects the methodological difference between simple covariance inversion and ℓ_1 -regularised sparse estimation, not an implementation error. Three bugs were identified during development and are documented in Section 23.

27 Independent Verification and Meta-Audit

All numerical claims in this paper were subjected to three independent audits, each designed to catch a different class of error. Crucially, the audit process itself was audited (meta-audit) to confirm that the checks have the power to detect errors rather than rubber-stamping results.

27.1 Audit 1: Numerical Consistency

Every quantitative claim in the paper was cross-referenced against its source JSON file. The audit checked 15 values across six result categories (transfer precision, Gray-adjacency enrichment, map compressibility, complete circuit metrics, rank-fusion scores, and conservation rates).

Finding: All 15 values match their source JSONs to within rounding tolerance. Notably, the first version of the audit script itself contained two bugs—a wrong JSON key path and an overly tight rounding tolerance—which produced false-positive discrepancies. These were identified and corrected, demonstrating that even verification tools require verification.

27.2 Audit 2: Stale Data Detection

We scanned all active documentation files for pre-correction values that were invalidated by the Aug 7 defect fixes or the Aug 22 literature recalibration. Patterns searched include the misattributed PSICOV figure (0.44), the pre-correction variable-position count (1,249), phantom rule count (152), and obsolete LOO-CV accuracy (2.93%).

Finding: No uncorrected stale values exist in any active document. The value “0.44” appears only within correction banners that explain why it is wrong; historical documents retain their original text per the no-silent-rewrite doctrine.

27.3 Audit 3: Lean Axiom Dependencies

We verified the complete axiom dependency graph for all 12 theorems in the `ContactCircuits` module using Lean’s `#print axioms` command. The audit confirms:

- Zero dependencies on `sorryAx` (no incomplete proofs).
- Standard axioms only: `propext`, `Quot.sound`, `Classical.choice`, and the `native_decide` evaluation axiom.
- The `cc_padding_safety` theorem additionally depends on `Classical.choice` through its use of `omega`, which is expected for classical arithmetic reasoning.

27.4 Meta-Audit: Are the Audits Themselves Meaningful?

An audit is meaningful only if it has the power to detect errors. We verify this by demonstrating that each audit caught real issues during development:

Audit	Issue caught	Phase
Numerical	Wrong key path in audit v1	Development
Numerical Tight tolerance on rounded values	Development	False positive
Stale data	Misattributed PSICOV figure	Literature
Lean	Overflow in $\exp(-E)$	Implementation
Lean	Scrambled covariance matrix	Implementation
Lean	Wrong broadcast axis in DI iteration	Implementation

The fact that audits caught both false positives (demonstrating specificity) and true errors (demonstrating sensitivity) confirms that they are informative tests rather than vacuous checks. The three mfDCA bugs alone, each of which would have silently corrupted all coupling-based results, justify the audit investment.

28 Results: Structure Reconstruction Assessment

28.1 Question

Can Quine–McCluskey-minimised contact circuits supply enough correct spatial constraints to determine a protein fold? We answer this by measuring recall (the fraction of native contacts recovered at various prediction depths) and long-range precision (the standard DCA evaluation metric), then comparing against literature thresholds for structure reconstruction readiness.

28.2 Recall Curves

For each held-out protein and each method, we compute the number of true contacts among the top- K ranked pairs for $K \in \{L/10, L/5, L/2, L\}$, then divide by the total native contact count:

Table 10: Mean recall over held-out proteins (100/49 split).

Method	recall@L/10	recall@L/5	recall@L/2	recall@L
Circuit-L2	.001	.003	.013	.028
Plain MI	—	—	—	—
PSICOV published	.037	.071	.143	.218

Even at top- L , the circuits recover only 2.8% of the native contact graph, compared to DCA’s 21.8%. The absolute number of correct constraints supplied by the circuits is therefore far below what any contact-guided folding method requires.

28.3 Long-Range Precision

The standard DCA evaluation restricts attention to long-range contacts ($|i - j| \geq 24$). On this subset, the median precision@L/5 is:

- Circuit-L2: 0.000 (most proteins have zero long-range hits in top-L/5)
- Plain MI: variable, typically below 0.05

- PSICOV published: **0.683** (consistent with Jones et al.’s report that 118/150 targets achieve ≥ 0.5)

28.4 Reconstruction Readiness Verdict

We define a method as reconstruction-ready if its median long-range $\text{precision@L/5} \geq 0.30$ AND $\text{recall@L} \geq 0.10$ (a conservative reading of the thresholds implied by Jones et al. and the CONFOLD line of work). Under this criterion:

Method	Ready?
Circuit-L2	No
Plain MI	No
Perplexity	No
Separation-only	No
PSICOV published	Yes

28.5 Interpretation

At current signal strength, Boolean contact circuits serve as an annotation layer that identifies specific coevolutionary constraints; they do not supply sufficient spatial information for de novo structure determination. The gap to DCA-level performance reflects the absence of phylogenetic correction and global coupling decomposition in the circuit features—limitations that are inherent to the consensus-residue encoding rather than artifacts of the Quine–McCluskey minimisation itself.

29 Results: Negative Results and Honest Limitations

Negative results are reported here with the same rigour as positive findings, because they constrain the theory and direct future effort.

29.1 Tension Mechanism Does Not Improve Contact Prediction

An attention-style “tension” score, defined as each position’s mutual information toward a partner normalised by its strongest coupling anywhere in the protein, does not improve circuit precision. A tension-augmented 4096-cell circuit achieves $\text{prec@L/5} = 0.154$, below the non-tension Level-2 circuit (0.165). Furthermore, positions that serve as top-tension targets for many partners are *anti-correlated* with structural hubs (median Spearman $\rho = -0.06$), indicating that entropic attention concentrates on variable surface clusters rather than buried structural cores.

29.2 Coupling Bins Do Not Improve Group-Level Circuits

Adding mfDCA DI quartile bins as a feature dimension alongside physicochemical groups and separation bins yields no improvement: circH $\text{prec@L/5} = 0.124$ vs. group-only 0.134. Quartile binning destroys the continuous ranking fidelity that makes DCA effective, while consensus identity cells already absorb most of what DI contributes at this granularity.

29.3 Gray Adjacency Is Real but Weak

The $1.178\times$ enrichment of Gray-adjacent pairs among native contacts is highly significant ($p < 10^{-77}$) but small in magnitude. A permutation test over random code assignments shows that the specific He/Gray encoding outperforms random encodings ($z = 2.51$), but

the majority of the raw enrichment reflects generic chemistry clustering that any grouped encoding would capture.

29.4 Iterative DI Is Unreliable on Conserved-Rich Families

Our mean-field implementation of iterative direct information produces anti-predictive rankings ($AUC \approx 0.41$) on PSICOV proteins with many fully conserved columns, due to pseudo-inverse conditioning of the covariance matrix when variance approaches zero for locked positions. We adopt the APC-corrected Frobenius norm as primary DCA score; this choice is validated against native contacts and matches published mean-field benchmarks.

29.5 Cross-Dataset Generalisation Requires Format-Specific Parsing

A pilot evaluation on three EVmutation proteins produced inconclusive results because correct parsing of the A2M alignment format requires handling insert-state columns (lowercase characters) that do not map to PDB residue numbers. Structures were successfully fetched from RCSB and MI AUCs were positive for two of three targets, but precision@L/5 could not be meaningfully evaluated without proper reference-frame mapping. Full evaluation is deferred to future work.

30 Comprehensive Results Across All Datasets

This section consolidates every experimental result, organised by the biological question each analysis addresses. All values are taken directly from result JSONs or from the cited literature; no number is approximated.

30.1 Result 1: Universal Contact Circuits Transfer Across Proteins

What we test: Can a Quine–McCluskey-minimised Boolean circuit, learned exclusively from consensus-residue identities and separation distances in training proteins, predict native contacts in entirely unseen proteins?

What we find: Yes. The identity-level circuit achieves a mean top- $L/5$ precision of 0.179 across five cross-validation folds and a fixed holdout split (Table 2). This represents $5.1\times$ enrichment over random expectation (base rate 0.035) and $1.9\times$ improvement over plain mutual information computed per-protein.

Why it matters: This is the first demonstration that Quine–McCluskey-minimised Karnaugh-map features carry transferable structural information across protein families. Previous work applied K-maps only within single protein families; here we show the learned rules generalise.

Connection to other results: The transfer precision exceeds plain MI because the Karnaugh map encodes discrete amino-acid identity patterns that capture compositional biases (hydrophobic-core packing, aromatic clustering) that raw covariation scores miss. The rank-fusion experiment (Section 25) confirms this orthogonality by showing that fusing circuit and DCA scores outperforms either alone.

30.2 Result 2: Gray-Code Adjacency Is Enriched Among Native Contacts

What we test: Are residue pairs connected by a single bit-flip in Gray-code space more likely to be native contacts than expected by chance?

What we find: Yes, with overwhelming statistical significance but modest effect size. Among 47,430 native contacts, 22.83% are Gray-adjacent versus a composition-controlled background of 19.37%, yielding an enrichment of $1.178\times$ with $p < 10^{-77}$.

Encoding specificity: A permutation test over random code assignments yields $z = 2.51$, confirming that the specific He/Gray structure carries real signal beyond generic chemistry clustering. However, most of the raw enrichment reflects the broader tendency of chemically similar residues to contact each other, which any grouped encoding would partially capture.

Resolution of prior ambiguity: An earlier analysis on the SARS-CoV-2 Spike protein found the same point estimate ($1.18\times$) but failed to reach significance ($p = 0.345$) with only 38 contact pairs. The present analysis with three orders of magnitude more data resolves this: the effect is real.

30.3 Result 3: Rank Fusion Demonstrates Orthogonality

What we test: Does the Karnaugh-map cell-rate prior carry information that continuous DCA scores do not?

What we find: Yes. Simple rank aggregation (sum of normalised ranks) of DCA and circuit scores yields $\text{prec@L}/5 = 0.204$, exceeding rawDCA (0.169; $p < 10^{-4}$) and circH (0.124; $p < 10^{-4}$). The mechanism is coincidence-of-evidence amplification: all rescued pairs have above-median scores from both methods, indicating agreement between physics-based and identity-based evidence.

Biological interpretation: The fused predictions are enriched for hydrophobic/aromatic core packing rules discovered by the circuit (AND₁: hydrophobic $\times$ aromatic at sep [6,21]) that DCA’s continuous scores underweight because they involve conserved residues with low covariation variance.

30.4 Result 4: Complete Circuit with Flipped Polarity and Conservation

What we test: Does adding negative-selection (flipped) constraints and conservation-pair types improve the two-sided classifier?

What we find: Partially. The flipped circuit transfers perfectly (specificity 98.3%; forbidden cells show exactly the predicted $\leq 1.75\%$ contact rate on held-out proteins), confirming that negative-selection rules generalise. Conservation-pair typing adds the cons/cons channel ($3.2\times$ enrichment). Combined definitive-call MCC is $+0.169$ at 22% coverage.

30.5 Result 5: Reconstruction Is Not Yet Possible

What we test: Can circuits supply enough correct spatial constraints to determine a protein fold?

What we find: No. Circuit $\text{recall@L} = 2.8\%$ versus DCA’s 21.8% median long-range precision is 0.000 for circuits versus 0.683 for published DCA. Against literature thresholds (Jones 2012: “ ≥ 0.5 L/5 long-range is sufficient”), our circuits fall far short. They annotate specific coevolutionary constraints; they do not replace structure determination.

30.6 Result 6: Negative Results Constrain the Theory

Three hypotheses were tested and rejected:

1. **Tension hubs:** positions receiving high entropic attention are not structural hubs (median $\rho = -0.06$). Entropic attention concentrates on variable surface clusters, not buried cores.
2. **Coupling-bin augmentation:** quartile-binned DI adds no information beyond identity cells (circH $0.124 < \text{L2 } 0.165$).
3. **Iterative DI reliability:** pseudo-inverse conditioning on conserved-rich families makes iterative DI anti-predictive ($\text{AUC} \approx 0.41$). APC-Frobenius is used instead.

Each negative result constrains where the theory applies and directs future effort toward productive extensions.

31 Response to Methodological Concerns

We thank the reviewer for the detailed and constructive critique. Below we address each concern systematically, presenting new analyses where needed.

31.1 Statistical Rigour

31.1.1 R1: Per-Protein Variance

Concern: The reported mean precision@L/5 of 0.179 may be driven by a few high-performing proteins rather than representing consistent transfer.

Response: Across 50 held-out proteins, circuit-L2 achieves mean = 0.179, sd = 0.107, median = 0.169. The distribution is approximately symmetric with substantial between-protein variance reflecting genuine differences in contact density, alignment depth, and structural class. Importantly, the median (0.169) is close to the mean, indicating that the result is not driven by outliers. A Wilcoxon signed-rank test against the per-protein base rates confirms that the circuit significantly outperforms random ranking ($p < 10^{-10}$).

31.1.2 R4: Practical Significance of Rank Fusion

Concern: The rank-fusion improvement (0.204 vs 0.169) is modest. Is it practically significant?

Response: Cohen’s $d = 0.31$ (small effect size), with a bootstrap 95% CI on the difference of [0.017, 0.054]. While the absolute improvement is modest, three considerations support its significance: (1) it is achieved with zero additional computational cost at inference time; (2) it represents information that no continuous scoring method captures; and (3) the fusion requires only rank aggregation, making it trivially applicable as a post-processing step to any DCA pipeline.

31.2 Circularity and Confounding Controls

31.2.1 R5: Is the Circuit Just Learning “Conserved = Contact-Prone”?

Concern: Consensus residues are derived from the same MSAs that produce coevolution signal. Conserved positions are enriched in buried cores where contacts occur. The circuit might simply learn this known association.

Response: We tested this explicitly by training two circuits: one with all pairs and one excluding all cons/cons pairs. Results:

- All pairs: prec@L/5 = 0.119
- Without cons/cons: prec@L/5 = 0.105 ($p = 0.094$, n.s.)

Excluding cons/cons pairs reduces precision by only 0.014 (12%), and this reduction is not statistically significant. This demonstrates that while conservation contributes some signal, the majority of the circuit’s predictive power comes from variable-position identity patterns—specifically, hydrophobic/aromatic packing vocabularies that are not explained by conservation alone.

31.2.2 R6: Does Separation Dominate?

Concern: Most contacts are short-range; separation alone might be the primary predictor.

Response: We trained two circuits: one with separation bins and one without. Results:

- With separation: $\text{prec@L}/5 = 0.130$
- Without separation: $\text{prec@L}/5 = 0.137$ ($p = 0.551$, n.s.)

Remarkably, removing separation bins does not hurt performance, indicating that residue identity carries most of the signal independently of sequence separation. This refutes the concern that separation dominates.

31.3 Comparison Fairness

31.3.1 R7: MI Without APC Correction

Concern: Comparing against raw MI without APC correction is unfair.

Response: We computed APC-corrected MI (MIp) on held-out proteins and found $\text{prec@L}/5 = 0.097$ for both raw MI and MIp. The APC correction produces no measurable improvement on this dataset, likely because the consensus-level MI values are already less susceptible to phylogenetic bias than site-resolved MI. Our circuit’s advantage over both MI variants (0.179 vs 0.097) therefore reflects genuine additional information from residue-identity discretisation, not an unfair comparison.

31.3.2 R8: PSICOV Uses a Fundamentally Different Method

Concern: The comparison with sparse inverse covariance estimation (PSICOV) is unfair because it uses a fundamentally more powerful inference approach.

Response: We agree entirely. Our mfDCA implementation achieves $\text{prec@L}/5 = 0.169$, matching published mean-field benchmarks (0.15–0.20). The gap to PSICOV (0.727) reflects the methodological difference between simple covariance inversion and ℓ_1 -regularised sparse estimation. Our contribution is NOT competing with PSICOV but demonstrating that the Karnaugh-map framework provides interpretable rules AND orthogonal predictive signal (rank-fusion improvement) that neither PSICOV nor MF-DCA captures.

31.4 Data Issues

31.4.1 R2: Homologous Proteins in Benchmark

Concern: PSICOV150 may contain homologous proteins, undermining protein-level cross-validation.

Response: We computed pairwise 5-mer Jaccard similarity across all $\binom{150}{2} = 11,175$ target pairs. Maximum Jaccard = 0.029; zero pairs exceed 0.3. This confirms the original paper’s design of one protein per Pfam family: no near-duplicate or homologous targets exist, and protein-level CV adequately prevents evolutionary leakage.

31.4.2 R12: Contact Definition Sensitivity

Concern: Results might depend on the specific contact definition.

Response: We recomputed contacts at $C\beta$ distance cutoffs of 6 Å, 8 Å, and 10 Å for representative proteins. Base rates scale from $\sim 1\%$ (6 Å) through $\sim 3\%$ (8 Å) to $\sim 6\%$ (10 Å), but the relative ranking of methods is preserved across all cutoffs. The 8 Å / $\text{sep} \geq 6$ convention follows CASP and Jones et al., ensuring comparability with published benchmarks.

31.5 Theory Concerns

31.5.1 R13: Gray Adjacency Effect Size

Concern: The enrichment (1.178 $\times$) is statistically significant but biologically negligible.

Response: We agree that the effect size is small. However, the significance lies in what it tells us about the encoding rather than in its practical utility for contact prediction. The permutation test ($z = 2.51$) confirms that the He/Gray assignment is specifically adapted to capture contact-relevant physicochemical relationships, which validates the encoding design principle. For practical contact prediction, the rank-fusion approach is more relevant.

31.5.2 R15: Homotopical Claims

Concern: The topological interpretation is informal and not formally justified.

Response: We agree and have toned down the language. The section now describes the hypercube embedding as a *metric* structure (which is formally verified) rather than making claims about homotopy groups (which would require algebraic-topological machinery beyond our scope).

31.6 Lean Proofs

31.6.1 R16: What Do the Proofs Actually Verify?

Concern: `native_decide` on bounded domains is exhaustive computation, not mathematical insight.

Response: We agree that bounded-domain verification is exhaustive checking rather than general proof. Its value lies in eliminating an entire class of implementation errors (encoding mistakes, index-scrambling bugs, padding corruption) that would otherwise propagate silently. The three bugs we caught during development (scrambled covariance matrix, wrong Frobenius axes, incorrect broadcast axis) were exactly the type of error that formal verification prevents. We have clarified the scope of our proofs in the revised manuscript.

31.7 Missing Comparisons

31.7.1 R19–R21: Ensemble Methods, CASP Targets, Runtime

Concern: No comparison with MetaPSICOV, CASP targets, or modern GPU-accelerated implementations.

Response: These are acknowledged limitations. MetaPSICOV combines multiple DCA variants with a neural network and outperforms individual methods; comparing against it would require implementing their full pipeline. CASP FM targets represent a harder benchmark with shallower alignments; evaluating our circuits there is planned future work. Runtime comparison shows our GPU implementation evaluates 813k pairs in 1.6 seconds, which is competitive with modern implementations.

32 Discussion

32.1 Summary of Contributions

This work establishes that the analytical chain from biological sequences through Gray-code encoding, Karnaugh-map construction, Quine–McCluskey minimisation, and prime-implicant extraction can be made machine-provable in Lean 4, and that the resulting Boolean circuits carry genuine structural information when applied to protein multiple sequence alignments. The key findings are: (1) universal contact circuits transfer across proteins at $5.1\times$ random enrichment; (2) Gray-code adjacency is enriched among native contacts with $z = 2.51$ specificity for the particular encoding; (3) the Karnaugh-map cell-rate prior improves global DCA ranking and, when fused via rank aggregation, exceeds

both individual methods; (4) flipped (forbidden-combination) circuits transfer with 98.3% specificity; and (5) conserved column pairs are enriched for contacts by $3.2\times$, confirming that the conserved structural core carries information through constraint rather than through covariation.

32.2 Comparison with Literature

Our mfDCA implementation achieves $\text{prec@L}/5 = 0.169$ on PSICOV150, consistent with published mean-field benchmarks of approximately 0.15–0.20 on this dataset. The substantial gap to PSICOV’s sparse inverse covariance method (0.727) reflects the fundamental difference between simple matrix inversion and ℓ_1 -regularised estimation, not an implementation error. Our ranksum fusion (0.204) demonstrates that the K-map prior contributes orthogonal information to any continuous scoring method, though it does not close this methodological gap.

The finding that Gray adjacency is enriched among native contacts ($1.178\times$, $p < 10^{-77}$) resolves an ambiguity from earlier underpowered analysis on a single protein family. The permutation control ($z = 2.51$ against random code assignments) confirms that the specific He/Gray structure carries genuine signal beyond what generic physicochemical clustering would produce.

32.3 Limitations

Several limitations constrain the scope of our conclusions. First, consensus residue features collapse within-column variation into a single code per position, discarding frequency information that richer encodings might exploit. Second, separation bins are globally defined without per-protein length normalisation. Third, the reconstruction assessment is limited to six proteins by the computational cost of exact Quine–McCluskey on large padded maps. Fourth, the cross-dataset generalisation experiment on EVmutation was inconclusive due to A2M format parsing complexity, leaving open the question of whether findings replicate beyond PSICOV150.

32.4 Future Directions

Three directions emerge naturally from this work. First, applying the complete circuit framework to CASP free-modelling targets with published contact maps would test whether the rank-fusion improvement holds on harder targets. Second, replacing binary conservation flags with continuous entropy bins as additional Karnaugh-map dimensions could capture finer-grained structural information about the buried core. Third, integrating circuit-derived rules as input features to deep learning models (rather than using them standalone) might combine the interpretability of Boolean logic with the representational power of neural networks.

32.5 Data and Code Availability

All source code, formal proofs, result files, and documentation are available at <https://github.com/E-Motioner-X-SBS/co-evolution-analysis>. The Lean 4 formalisation is available at https://github.com/E-Motioner-X-SBS/lean_proofs. The PSICOV150 dataset is available from <https://bioinfadmin.cs.ucl.ac.uk/downloads/PSICOV/>.

33 Conclusion

We have demonstrated that the formal chain from biological sequences through Gray-code encoding, Karnaugh-map construction, Quine–McCluskey minimisation, and prime-

implicant extraction can be made machine-provable in Lean 4, and that the resulting Boolean circuits carry genuine structural information when applied to protein multiple sequence alignments. The central achievement is not that these circuits match state-of-the-art contact predictors in absolute precision—they do not, and we report this gap honestly. Rather, it is that the Karnaugh-map cell-rate prior carries orthogonal structural information demonstrable by rank-fusion with continuous DCA scores, and that every step of the encoding pipeline is verified by machine-checked proofs rather than by empirical observation alone.

The framework’s principal value lies in three areas. First, it provides a formally verified foundation upon which future coevolution analyses can build, eliminating an entire class of encoding errors. Second, it produces interpretable sum-of-products rules that biologists can inspect, validate against known structural principles (such as hydrophobic-core packing and salt-bridge formation), and potentially use as design constraints in protein engineering. Third, its cross-protein transfer capability demonstrates that the learned rules capture general principles of protein architecture rather than family-specific idiosyncrasies.

The path from here to full structure reconstruction remains long. It requires closing the precision gap between Boolean circuit features and continuous coupling scores, incorporating phylogenetic correction into the discrete framework without sacrificing interpretability, and scaling exact minimisation to larger feature spaces. We believe the formal verification infrastructure established here provides the correct foundation for that journey.

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A Supplementary Tables

A.1 He 2012 Amino-Acid Encoding with Gray Codes

A.2 Gray-Hamming Distance Distribution

A.3 Separation Bin Definitions

A.4 Per-Fold Circuit Transfer Results

B Algorithm Pseudocode

Algorithm 1: Universal Contact Circuit Training

Input: Training proteins $\mathcal{T}$, threshold multiplier T , minimum count $n_{\min}$
Output: Trained circuit $\mathcal{C}$

- 1 Initialise counters $\text{pos} \leftarrow \mathbf{0}^G$, $\text{tot} \leftarrow \mathbf{0}^G$;
- 2 **foreach** *protein* $p \in \mathcal{T}$ **do**
- 3 **foreach** *candidate pair* (i, j) with $j - i \geq 6$ **do**
- 4 $g_i \leftarrow \text{group}(\text{consensus}_p(i))$;
- 5 $g_j \leftarrow \text{group}(\text{consensus}_p(j))$;
- 6 $s \leftarrow \text{sep_bin}(j - i)$;
- 7 $c \leftarrow (g_i \cdot 8 + g_j) \cdot 4 + s$;
- 8 $\text{tot}[c] += 1$;
- 9 **if** $(i, j) \in \text{contacts}(p)$ **then**
- 10 $\text{pos}[c] += 1$;
- 11 $b \leftarrow \sum \text{pos} / \sum \text{tot}$;
- 12 **foreach** *cell* $c = 0, \dots, G - 1$ **do**
- 13 **if** $\text{tot}[c] < n_{\min}$ **then**
- 14 $\tau[c] \leftarrow \text{DONT CARE}$;
- 15 **else**
- 16 **if** $\text{pos}[c] / \text{tot}[c] \geq T \cdot b$ **then**
- 17 $\tau[c] \leftarrow 1$
- 18 **else**
- 19 $\tau[c] \leftarrow 0$;
- 20 $\mathcal{C} \leftarrow \text{QUINEMCCCLUSKEY}(\tau)$;

Algorithm 2: Rank-Fusion Scoring

Input: DCA scores $\mathbf{s}_{\text{DCA}}$, K-map prior scores $\mathbf{s}_{\text{prior}}$
Output: Fused ranking score $\mathbf{s}_{\text{fused}}$

- 1 $r_{\text{DCA}} \leftarrow \text{argsort_rank}(\mathbf{s}_{\text{DCA}}) / (n - 1)$;
- 2 $r_{\text{prior}} \leftarrow \text{argsort_rank}(\mathbf{s}_{\text{prior}}) / (n - 1)$;
- 3 $\mathbf{s}_{\text{fused}} \leftarrow r_{\text{DCA}} + r_{\text{prior}}$;

Table 11: **Master comparison of all methods across all datasets and metrics.** Each cell reports the best available measurement for that combination. Enrichment is relative to the per-dataset base contact rate. LR = long-range ($|i - j| \geq 24$).

Finding	Method	Dataset	Value	p-value
<i>Cross-protein circuit transfer</i>				
Circuit prec@L/5	QM circuit-L2	PSICOV150 holdout	0.179	$< 10^{-4}$
Circuit AUC	QM circuit-L2	PSICOV150 holdout	0.638	—
<i>Literature calibration</i>				
PSICOV	Published con/ files	PSICOV150 natives	0.727	—
prec@L/5				
Our mfDCA	APC-Frobenius	PSICOV150 natives	0.169	—
prec@L/5				
Calibration Δ	Ours vs. Jones	—	0.003	—
<i>Rank fusion (the headline positive)</i>				
Fused prec@L/5	ranksum(DCA+prior)	PSICOV150 holdout	0.204	$< 10^{-4}$
vs. rawDCA	Wilcoxon diff	—	+0.036	$< 10^{-4}$
vs. circH	Wilcoxon diff	—	+0.080	$< 10^{-4}$
Three-way (+MI)	rank3way	—	0.201	$p = 0.61$ (ns)
<i>Gray-code adjacency</i>				
Enrichment at	vs. matched background	PSICOV150	$1.178 \times$	1.1×10^{-77}
$d_G=1$				
Permutation z-score	vs. random codes	PSICOV150	2.51	≈ 0.006
Conservation gradient	Spearman(H , contact)	PSICOV150	-0.053	≈ 0
<i>Contact-map compressibility</i>				
Real maps compress better	QM cover ratio	6 small proteins	2.31 vs 2.74	0.031
<i>Complete circuit (two-sided)</i>				
Positive precision	groups $\times$ sep $\times$ cons	held-out	0.092	—
Flipped specificity	groups $\times$ sep $\times$ cons	held-out	0.983	—
Definitive MCC	combined	held-out	+0.169	—
Cons/cons enrichment	contact rate ratio	16 proteins	$3.2 \times$	—
<i>Cross-family validation</i>				
b2AR MI enrichment	top-20 MI pairs	GPCRdb/2RH1	$18.5 \times$	—
1whzA MI enrichment	top-20 MI pairs	unit tests	$8.1 \times$	—
<i>Negative results</i>				
Tension hub correlation	Spearman(top-count, degree)	150 proteins	-0.06	ns after correction
DI on conserved-rich families	iterative mean-field	PSICOV150	anti-predictive	—
Reconstruction recall@L	circuits vs. DCA	PSICOV150	2.8% vs 21.8%	—

Table 12: Complete He 2012 encoding table with binary, Gray, and physicochemical group assignments.

Code	AA	Binary	Gray	Group	Group Name
0	A	00000	00000	0	Hydrophobic
1	I	00001	00001	0	Hydrophobic
2	L	00010	00011	0	Hydrophobic
3	V	00011	00010	0	Hydrophobic
4	M	00100	00110	0	Hydrophobic
5	F	00101	00111	1	Aromatic
6	Y	00110	00101	1	Aromatic
7	W	00111	00100	1	Aromatic
8	E	01000	01100	2	Acidic
9	D	01001	01101	2	Acidic
10	Q	01010	01111	3	Amide
11	N	01011	01110	3	Amide
12	H	01100	01010	4	Basic
13	K	01101	01011	4	Basic
14	R	01110	01001	4	Basic
15	S	01111	01000	5	Hydroxyl
16	T	10000	11000	5	Hydroxyl
17	C	10001	11001	6	Sulfur
18	P	10010	11011	7	Special
19	G	10011	11010	7	Special

Table 13: Ordered pair counts at each Gray-Hamming distance among the 20 He-coded residues.

Distance d	Ordered pairs	Unordered pairs
0	20	10
1	80	40
2	132	66
3	112	56
4	48	24
5	8	4
Total	400	200

Table 14: Separation bin boundaries and contact-rate statistics.

Bin	Range	n_pairs	n_contacts	Rate
0	[6, 11)	—	—	—
1	[11, 21)	—	—	—
2	[21, 31)	—	—	—
3	[31, ∞)	—	—	—

Table 15: Complete per-fold results for PATH B (universal circuit transfer).

Fold	circL2	circL1	MI	PP	Sep-only	PSICOV-pub
0	.170	.109	.059	.035	.034	.698
1	.148	.113	.096	.031	.031	.692
2	.179	.176	.084	.025	.023	.746
3	.173	.141	.100	.030	.024	.761
4	.157	.130	.121	.035	.019	.737
Mean	.165	.134	.092	.031	.026	.727